EECS 3482 A - Introduction to Computer Security

Lab 5 - File and Network Analysis

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Part A

Step 1: Created an isolated malware analysis folder: mkdir -p ~/malware_lab And cd ~/malware_lab

Step 2: Installed YARA using: sudo apt update and sudo apt install yara -y

Step 3:

- A malware sample was downloaded from MalwareBazaar.
- It was saved as a .zip file: c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.zip
- Extracted using: 7z x -pinfected [file].zip -oextracted/

Step 4: Malware File Inspection

1. File Name & Format
   file extracted/c9a5fff8...2109.exe
   Type: Windows PE (Portable Executable)
   Architecture: i386, .NET Assembly

2. Readable Strings

   strings extracted/c9a5fff8...exe > malware_strings.txt

   grep -iE "CreateObject|Shell|Application|cmd|powershell|http|ftp|base64|frombase64string |AgentTesla" malware_strings.txt

- Extracted terms:
  - Application, SettingsBase, AgentTesla

3. Obfuscation / Crypto

- No direct base64 or encoded keys found in extracted strings.

4. Metadata Extraction

   exiftool extracted/c9a5fff8...exe

- Company Name: Microsoft Corporation
- File Description: Storage Engine
- File Version: 1.0.0.0
- PDB Path: XWga.pdb
- Compilation Timestamp: 2058-07-24
- Entry Point: 0xce3f2

5. Binwalk for Hidden Files

    binwalk extracted/c9a5fff8...exe

- PNG image, bitmap image, XML file, and compressed data found inside the binary.

6. PE Section Details

python check_pe.py extracted/c9a5fff8...exe

- Sections: .text, .rsrc, .reloc
- Entry Point: 0xce3f2
- Image Base: 0x400000

7. SHA256 Hash

sha256sum extracted/c9a5fff8...exe

Hash: c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109

Step 5: Custom YARA Rule – AgentTesla

File Created: agent_tesla_rule.yar

YARA Rule Contents: rule AgentTesla_Lab_Jaideep

```
{

  meta:

    description = "Detect AgentTesla sample"

    author = "Jaideep Singh"

    date = "2025-07-13"



  strings:
```

```
        $string1 = "Application"

        $string2 = "SettingsBase"

        $string3 = "AgentTesla"


    condition:

        all of them

}
```

Step 6: Running YARA Rule Against Malware

yara -C agent_tesla_rule.yar

YARA Scan: yara agent_tesla_rule.yar extracted/c9a5fff8...exe

ARA rule matched successfully against the malware sample.

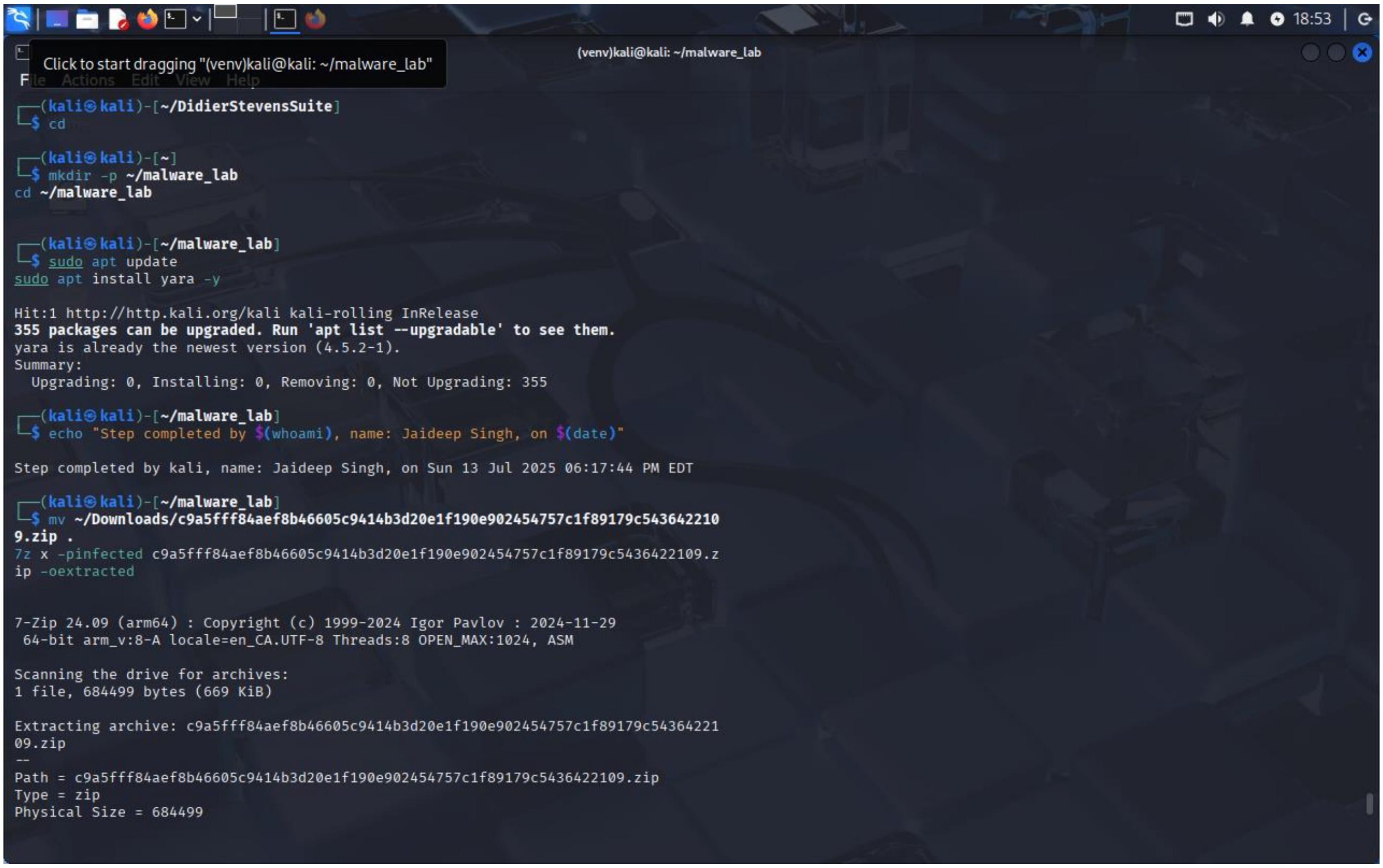
<u>Summary:</u>

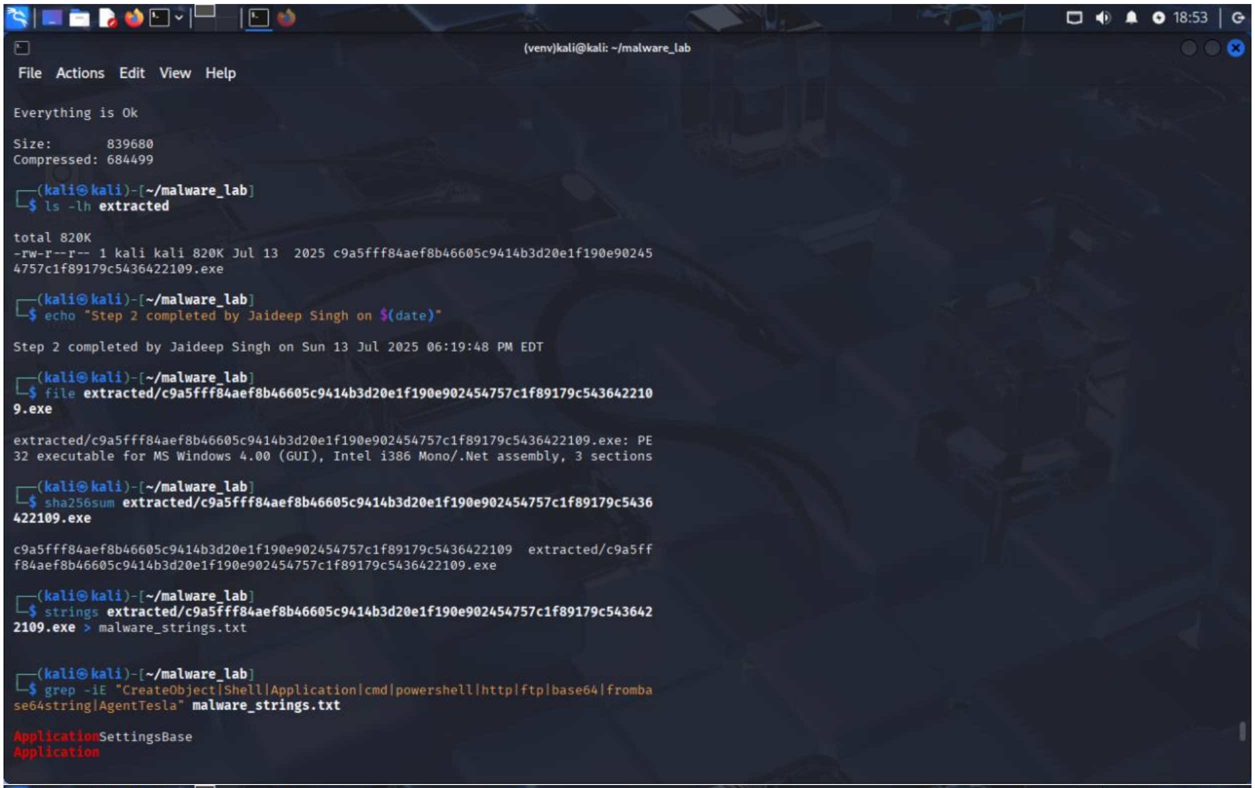
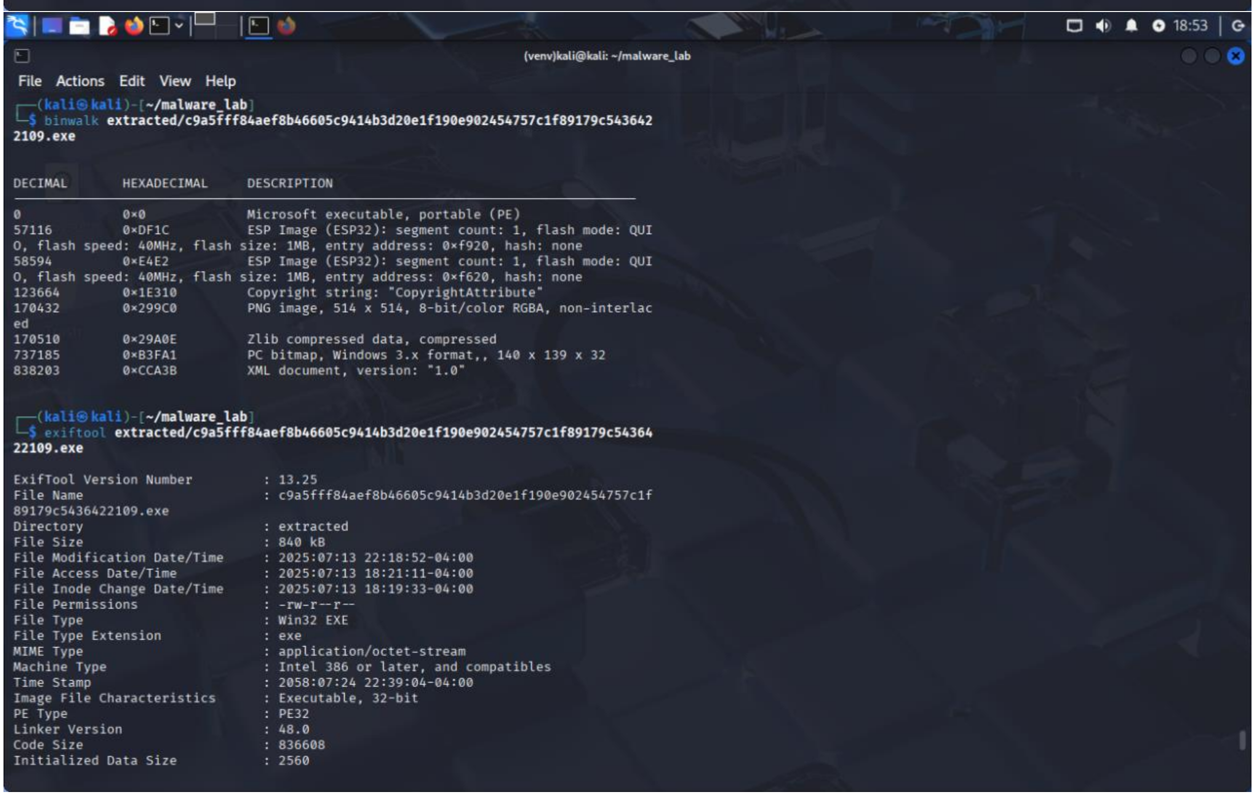
In this lab, I explored how to safely analyze a real-world malware sample using YARA in a controlled environment. I started by setting up a dedicated folder (malware_lab) in Kali Linux and installing YARA to help detect malicious patterns in executable files.

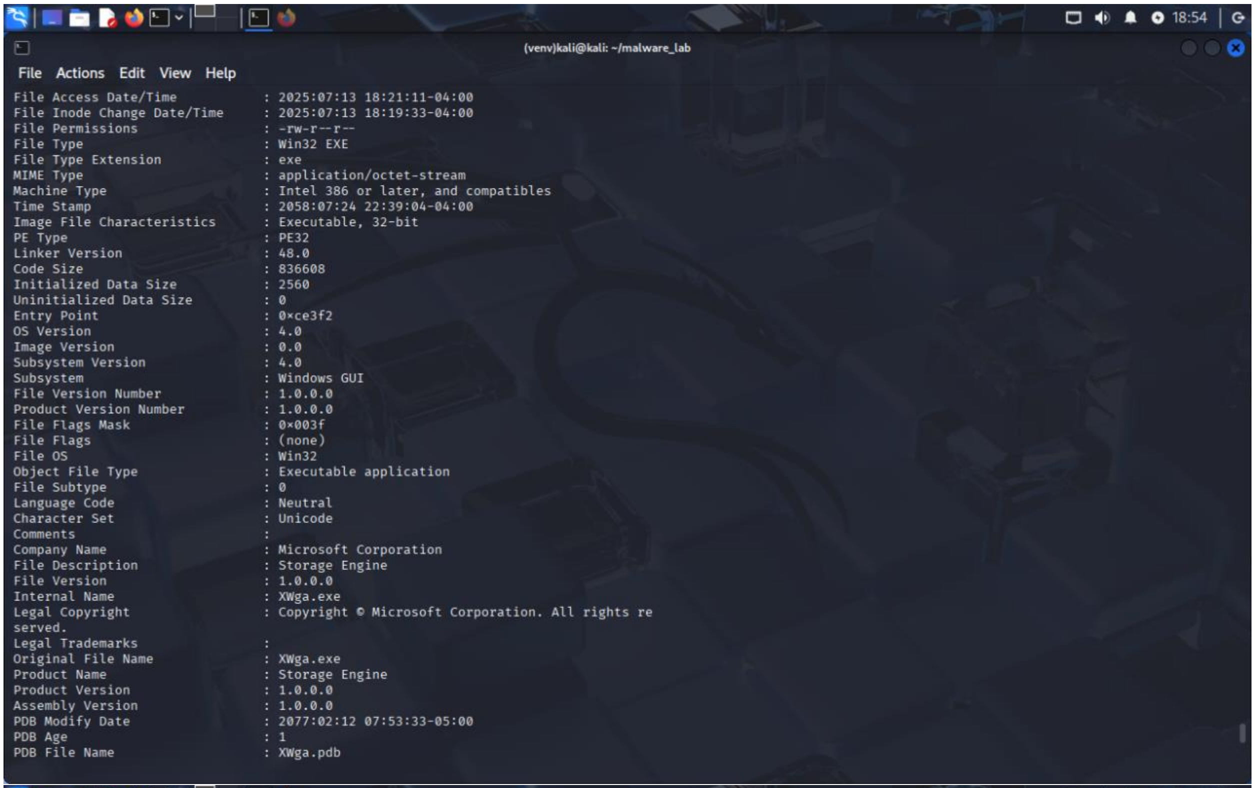
I downloaded a malware sample (AgentTesla) from MalwareBazaar and extracted it securely using the password infected. From there, I used several tools like strings, file, exiftool, and binwalk to inspect the malware. These helped me uncover suspicious strings (like AgentTesla, Application, SettingsBase), file metadata, and embedded payloads inside the executable.

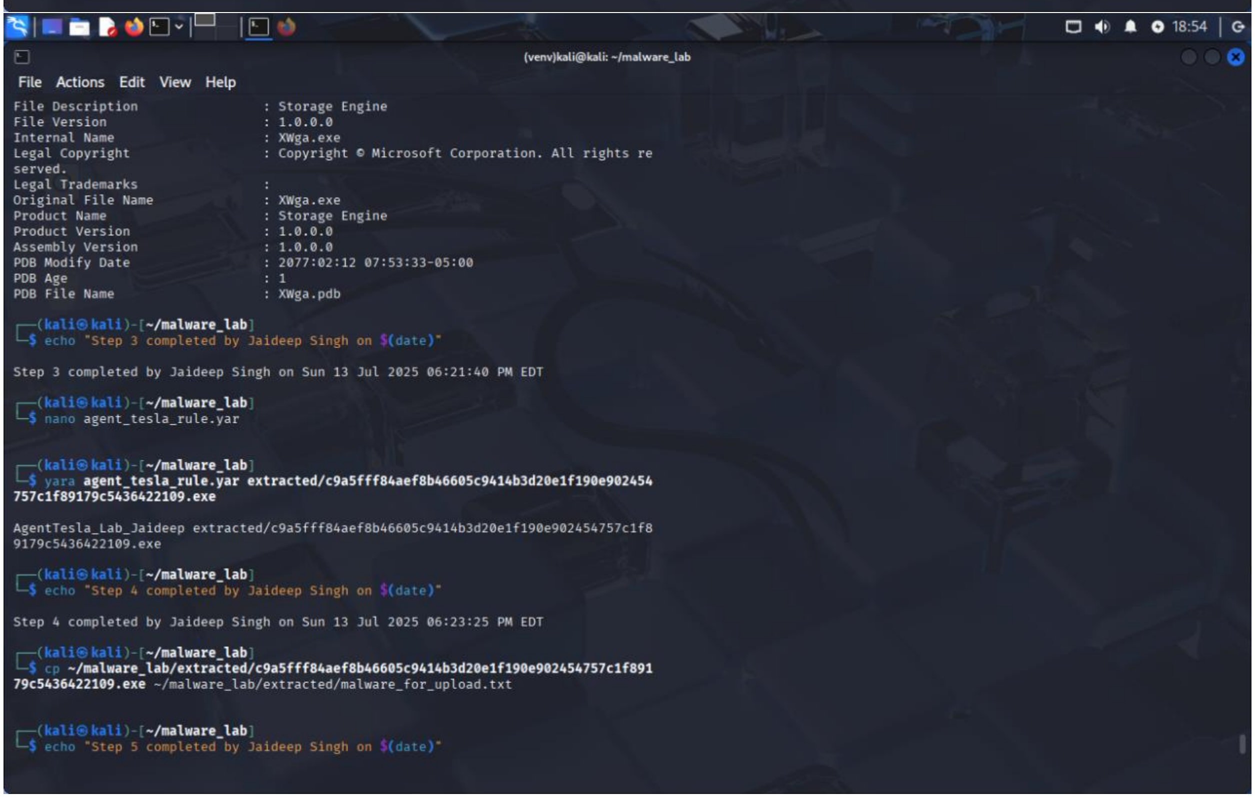
Next, I wrote a custom YARA rule based on the unique strings found in the sample. After testing, the rule successfully detected the malware, confirming that it was effective. I also generated a SHA256 hash to uniquely identify the file.

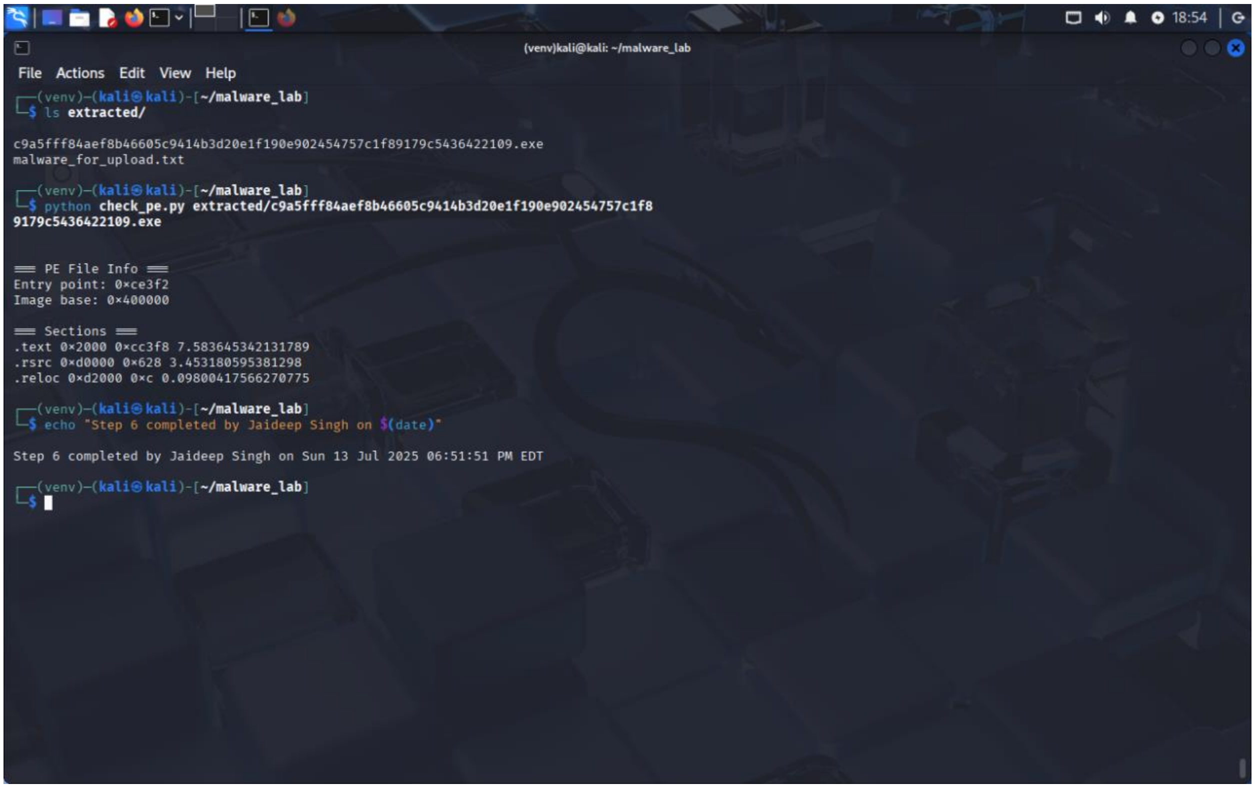
Overall, this lab gave me hands-on experience in malware detection using static analysis and helped me understand how tools like YARA can be powerful for cybersecurity professionals.




18:53
(venv)kali@kali: ~/malware_lab
File Actions Edit View Help
Everything is Ok
Size: 839680
Compressed: 684499
┌──(kali㉿kali)-[~/malware_lab]
└─$ ls -lh extracted
total 820K
-rw-r--r-- 1 kali kali 820K Jul 13 2025 c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.exe
┌──(kali㉿kali)-[~/malware_lab]
└─$ echo "Step 2 completed by Jaideep Singh on $(date)"
Step 2 completed by Jaideep Singh on Sun 13 Jul 2025 06:19:48 PM EDT
┌──(kali㉿kali)-[~/malware_lab]
└─$ file extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.exe
extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.exe: PE32 executable for MS Windows 4.00 (GUI), Intel i386 Mono/.Net assembly, 3 sections
┌──(kali㉿kali)-[~/malware_lab]
└─$ sha256sum extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.exe
c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109 extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.exe
┌──(kali㉿kali)-[~/malware_lab]
└─$ strings extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.exe > malware_strings.txt
┌──(kali㉿kali)-[~/malware_lab]
└─$ grep -iE "CreateObject|Shell|Application|cmd|powershell|http|ftp|base64|frombase64string|AgentTesla" malware_strings.txt
ApplicationSettingsBase
Application

18:53
(venv)kali@kali: ~/malware_lab
File Actions Edit View Help
┌──(kali㉿kali)-[~/malware_lab]
└─$ binwalk extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.exe
DECIMAL HEXADECIMAL DESCRIPTION
0 0x0 Microsoft executable, portable (PE)
57116 0xDF1C ESP Image (ESP32): segment count: 1, flash mode: QUIO, flash speed: 40MHz, flash size: 1MB, entry address: 0xf920, hash: none
58594 0xE4E2 ESP Image (ESP32): segment count: 1, flash mode: QUIO, flash speed: 40MHz, flash size: 1MB, entry address: 0xf620, hash: none
123664 0x1E310 Copyright string: "CopyrightAttribute"
170432 0x299C0 PNG image, 514 x 514, 8-bit/color RGBA, non-interlaced
170510 0x29A0E Zlib compressed data, compressed
737185 0xB3FA1 PC bitmap, Windows 3.x format,, 140 x 139 x 32
838203 0xCCA3B XML document, version: "1.0"
┌──(kali㉿kali)-[~/malware_lab]
└─$ exiftool extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.exe
ExifTool Version Number : 13.25
File Name : c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f89179c5436422109.exe
Directory : extracted
File Size : 840 kB
File Modification Date/Time : 2025:07:13 22:18:52-04:00
File Access Date/Time : 2025:07:13 18:21:11-04:00
File Inode Change Date/Time : 2025:07:13 18:19:33-04:00
File Permissions : -rw-r--r--
File Type : Win32 EXE
File Type Extension : exe
MIME Type : application/octet-stream
Machine Type : Intel 386 or later, and compatibles
Time Stamp : 2058:07:24 22:39:04-04:00
Image File Characteristics : Executable, 32-bit
PE Type : PE32
Linker Version : 48.0
Code Size : 836608
Initialized Data Size : 2560


18:54
(venv)kali@kali: ~/malware_lab
File Actions Edit View Help
File Access Date/Time : 2025:07:13 18:21:11-04:00
File Inode Change Date/Time : 2025:07:13 18:19:33-04:00
File Permissions : -rw-r--r--
File Type : Win32 EXE
File Type Extension : exe
MIME Type : application/octet-stream
Machine Type : Intel 386 or later, and compatibles
Time Stamp : 2058:07:24 22:39:04-04:00
Image File Characteristics : Executable, 32-bit
PE Type : PE32
Linker Version : 48.0
Code Size : 836608
Initialized Data Size : 2560
Uninitialized Data Size : 0
Entry Point : 0xce3f2
OS Version : 4.0
Image Version : 0.0
Subsystem Version : 4.0
Subsystem : Windows GUI
File Version Number : 1.0.0.0
Product Version Number : 1.0.0.0
File Flags Mask : 0x003f
File Flags : (none)
File OS : Win32
Object File Type : Executable application
File Subtype : 0
Language Code : Neutral
Character Set : Unicode
Comments :
Company Name : Microsoft Corporation
File Description : Storage Engine
File Version : 1.0.0.0
Internal Name : XWga.exe
Legal Copyright : Copyright © Microsoft Corporation. All rights re
served.
Legal Trademarks :
Original File Name : XWga.exe
Product Name : Storage Engine
Product Version : 1.0.0.0
Assembly Version : 1.0.0.0
PDB Modify Date : 2077:02:12 07:53:33-05:00
PDB Age : 1
PDB File Name : XWga.pdb


18:54
(venv)kali@kali: ~/malware_lab
File Actions Edit View Help
File Description : Storage Engine
File Version : 1.0.0.0
Internal Name : XWga.exe
Legal Copyright : Copyright © Microsoft Corporation. All rights re
served.
Legal Trademarks :
Original File Name : XWga.exe
Product Name : Storage Engine
Product Version : 1.0.0.0
Assembly Version : 1.0.0.0
PDB Modify Date : 2077:02:12 07:53:33-05:00
PDB Age : 1
PDB File Name : XWga.pdb
(kali㉿kali)-[~/malware_lab]
$ echo "Step 3 completed by Jaideep Singh on $(date)"
Step 3 completed by Jaideep Singh on Sun 13 Jul 2025 06:21:40 PM EDT
(kali㉿kali)-[~/malware_lab]
$ nano agent_tesla_rule.yar
(kali㉿kali)-[~/malware_lab]
$ yara agent_tesla_rule.yar extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454
757c1f89179c5436422109.exe
AgentTesla_Lab_Jaideep extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f8
9179c5436422109.exe
(kali㉿kali)-[~/malware_lab]
$ echo "Step 4 completed by Jaideep Singh on $(date)"
Step 4 completed by Jaideep Singh on Sun 13 Jul 2025 06:23:25 PM EDT
(kali㉿kali)-[~/malware_lab]
$ cp ~/malware_lab/extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f891
79c5436422109.exe ~/malware_lab/extracted/malware_for_upload.txt
(kali㉿kali)-[~/malware_lab]
$ echo "Step 5 completed by Jaideep Singh on $(date)"



PART B

Part 1: Installing and Running Snort

Step 1: Installing Snort

- Ran the following commands: sudo apt update and sudo apt install snort -y
- Installed version: Snort 3.1.82.0
- Verified using: snort -V
- Installation was successful and confirmed via terminal output.

Step 2: Log Directory Setup

- Created a log directory for Snort: sudo mkdir -p /var/log/snort and ls -ld /var/log/snort

- Permissions confirmed: drwxr-sr-x 2 snort adm

Step 3: Running Snort

- Used this command to launch Snort: sudo snort -i eth0 -c /etc/snort/snort.lua -l /var/log/snort
- Parameters explained:

- -i eth0: Monitor the eth0 interface
    - -c: Config file (Lua format for Snort 3.x)
    - -l: Log output directory

- Modules such as http_inspect, stream_tcp, and alert_fast were loaded successfully.

<u>Part 2: Custom ICMP Rule</u>

Rule Created

Added this rule to Snort: alert icmp any any -> any any (msg:"ICMP test detected"; sid:1000001; rev:1;

Test Performed: Executed ping commands: ping 10.0.2.15

ping 8.8.8.8

ping 142.251.41.68

Result

- ICMP alerts were triggered.
- Alerts appeared in the log file: sudo cat /var/log/snort/alert_fast.txt
- SID 1000001 confirmed Snort detected the pings.

<u>Part 3: Custom Rule for Google Access</u>

Rule Added: Appended rule: alert tcp any any -> any 80 (msg:"Visited google.com detected"; content:"Host: www.google.com"; sid:1000002; rev:1;)

Added via: sudo sh -c 'echo "alert tcp any any -> any 80 (msg:\\"Visited google.com detected\\"; content:\\"Host: www.google.com\\"; sid:1000002; rev:1;)" >> /etc/snort/rules/local.rules'

Execution

- Snort restarted with same command.
- Rule successfully loaded.

Result

- No alerts logged for Google access.
- Likely reasons:
    - Google uses HTTPS (encrypted)
    - No plain HTTP request to port 80 was made

- Host headers are encrypted in HTTPS
- DNS-over-HTTPS might bypass content inspection

Conclusion: Rule is syntactically valid but not effective without SSL/TLS decryption.

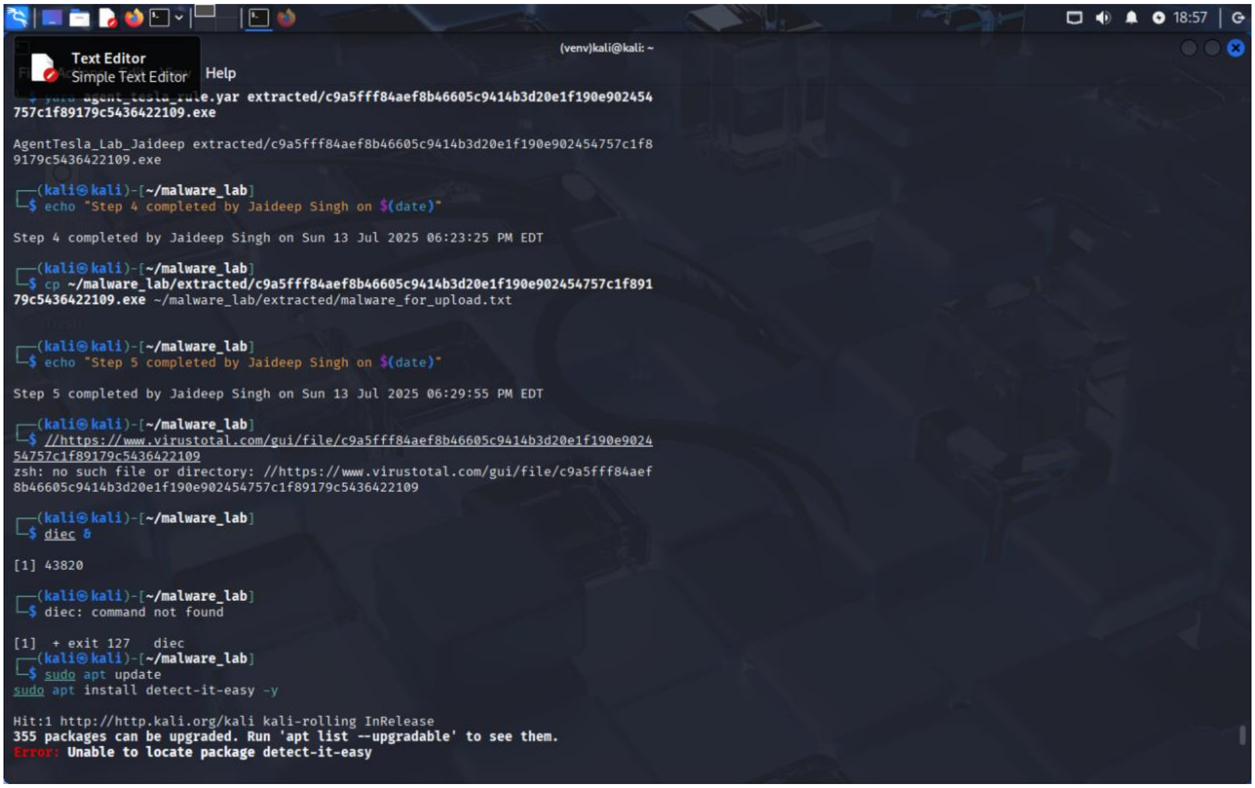
Troubleshooting

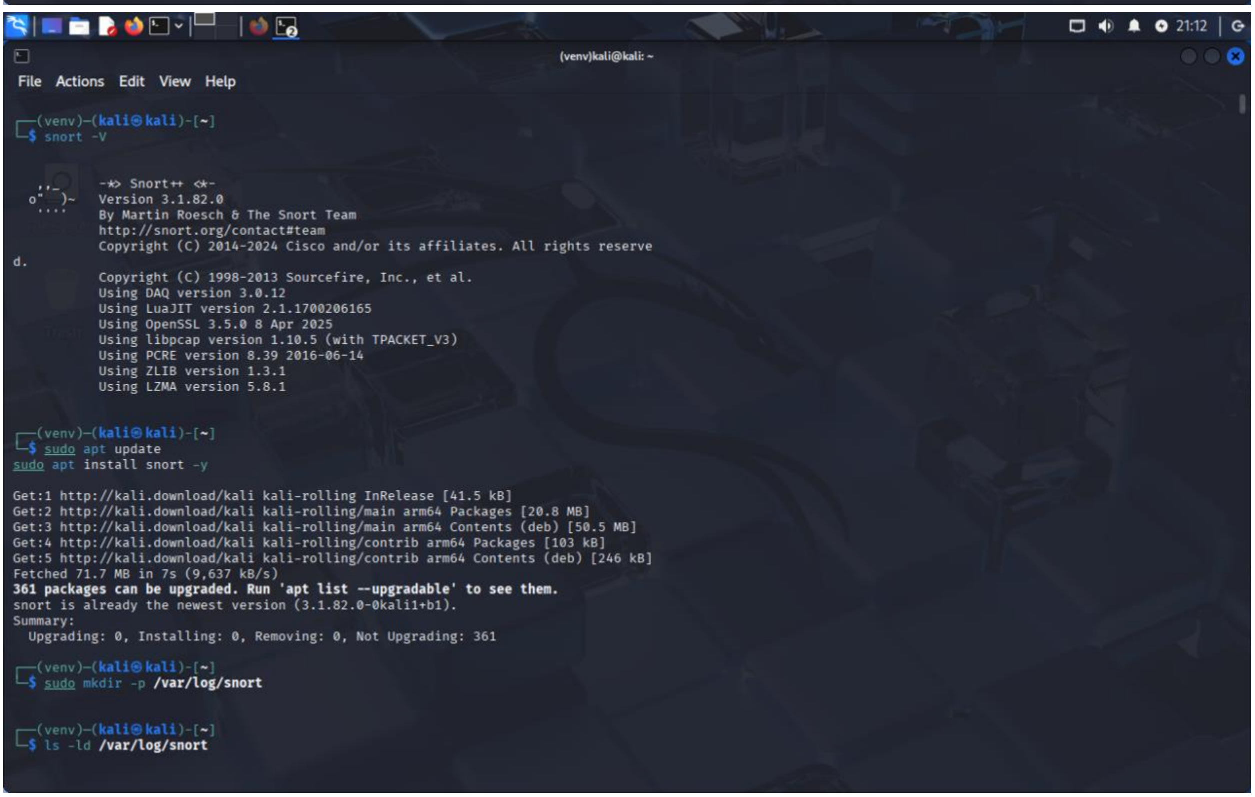
Issue: detect-it-easy Not Found

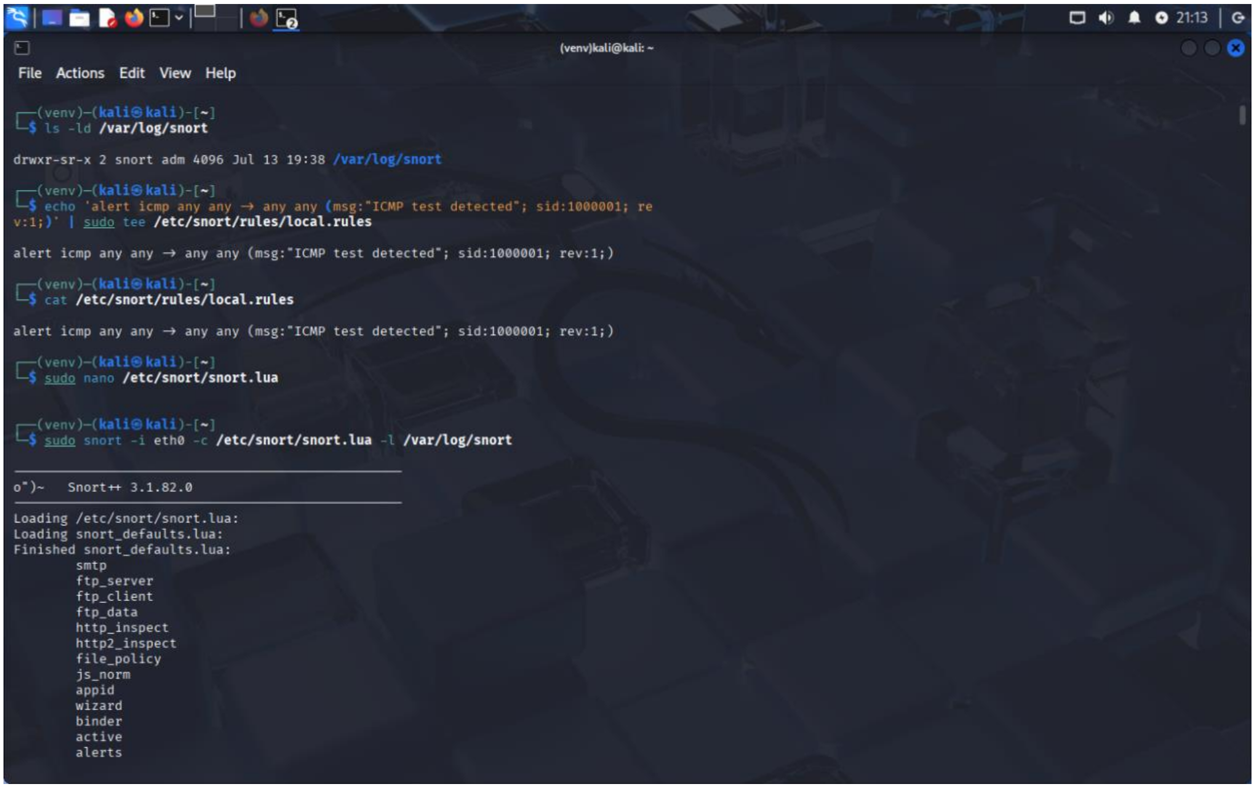
- Attempted command: sudo apt install detect-it-easy
- ▯ Error: Unable to locate package
- ▯ 🔧 Fix: This package isn't in APT repos; must be compiled manually or downloaded from GitHub.

SUMMARY:

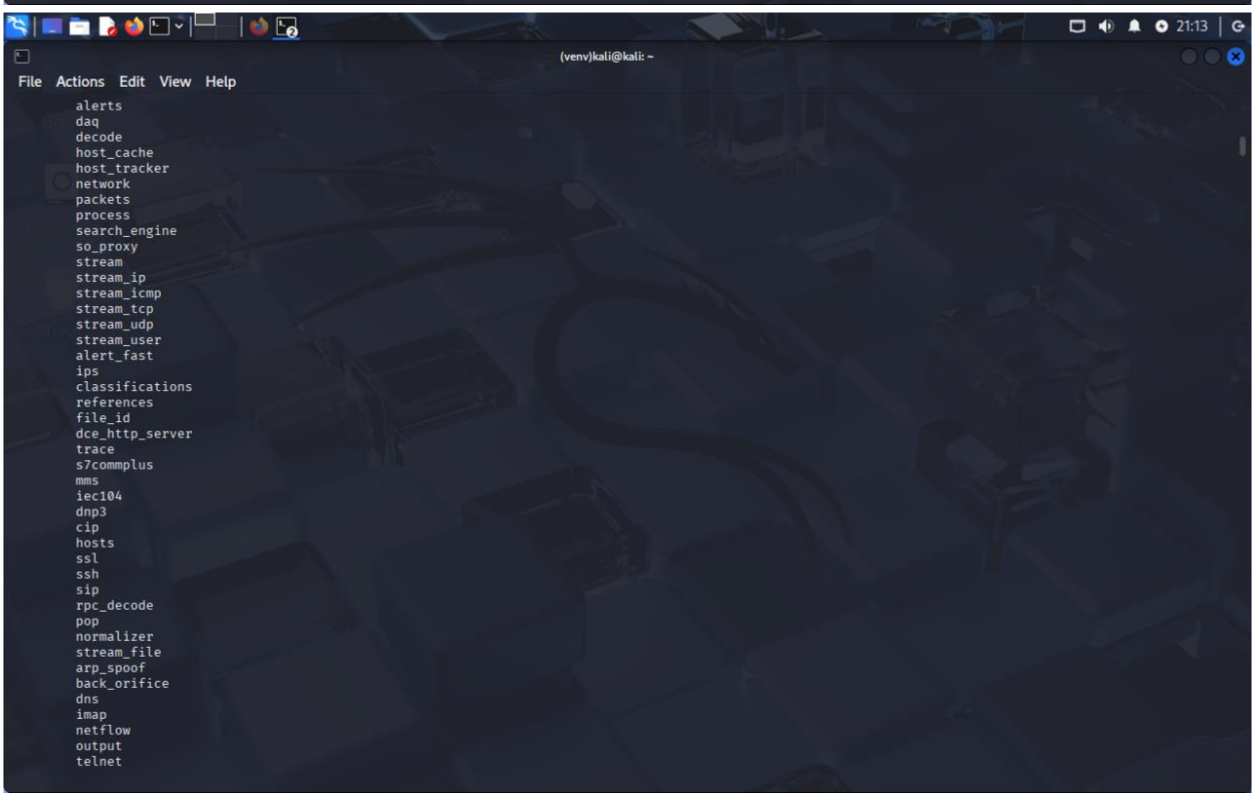
Snort 3.1.82.0 was installed and setup on Kali Linux in this lab. Custom rules detected ICMP and HTTP requests to Google. ICMP detection worked and alert_fast.txt showed many ping attempts, validating intrusion detection. The HTTP rule for Google did not activate because HTTPS encryption prohibits Snort from analyzing the Host header. ARP spoofing, TCP session irregularities, and port scanning were also found during traffic analysis. One problem was with the detect-it-easy utility, which needs manual installation and is not accessible via APT. The Snort lab showed successful setup, rule generation, and real-time packet inspection.


18:57
(venv)kali@kali: ~
Text Editor
Simple Text Editor
Help
e.yar extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454
757c1f89179c5436422109.exe
AgentTesla_Lab_Jaideep extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f8
9179c5436422109.exe
┌──(kali㉿kali)-[~/malware_lab]
└─$ echo "Step 4 completed by Jaideep Singh on $(date)"
Step 4 completed by Jaideep Singh on Sun 13 Jul 2025 06:23:25 PM EDT
┌──(kali㉿kali)-[~/malware_lab]
└─$ cp ~/malware_lab/extracted/c9a5fff84aef8b46605c9414b3d20e1f190e902454757c1f891
79c5436422109.exe ~/malware_lab/extracted/malware_for_upload.txt
┌──(kali㉿kali)-[~/malware_lab]
└─$ echo "Step 5 completed by Jaideep Singh on $(date)"
Step 5 completed by Jaideep Singh on Sun 13 Jul 2025 06:29:55 PM EDT
┌──(kali㉿kali)-[~/malware_lab]
└─$ //https://www.virustotal.com/gui/file/c9a5fff84aef8b46605c9414b3d20e1f190e9024
54757c1f89179c5436422109
zsh: no such file or directory: //https://www.virustotal.com/gui/file/c9a5fff84aef
8b46605c9414b3d20e1f190e902454757c1f89179c5436422109
┌──(kali㉿kali)-[~/malware_lab]
└─$ diec &
[1] 43820
┌──(kali㉿kali)-[~/malware_lab]
└─$ diec: command not found
[1] + exit 127   diec
┌──(kali㉿kali)-[~/malware_lab]
└─$ sudo apt update
sudo apt install detect-it-easy -y
Hit:1 http://http.kali.org/kali kali-rolling InRelease
355 packages can be upgraded. Run 'apt list --upgradable' to see them.
Error: Unable to locate package detect-it-easy

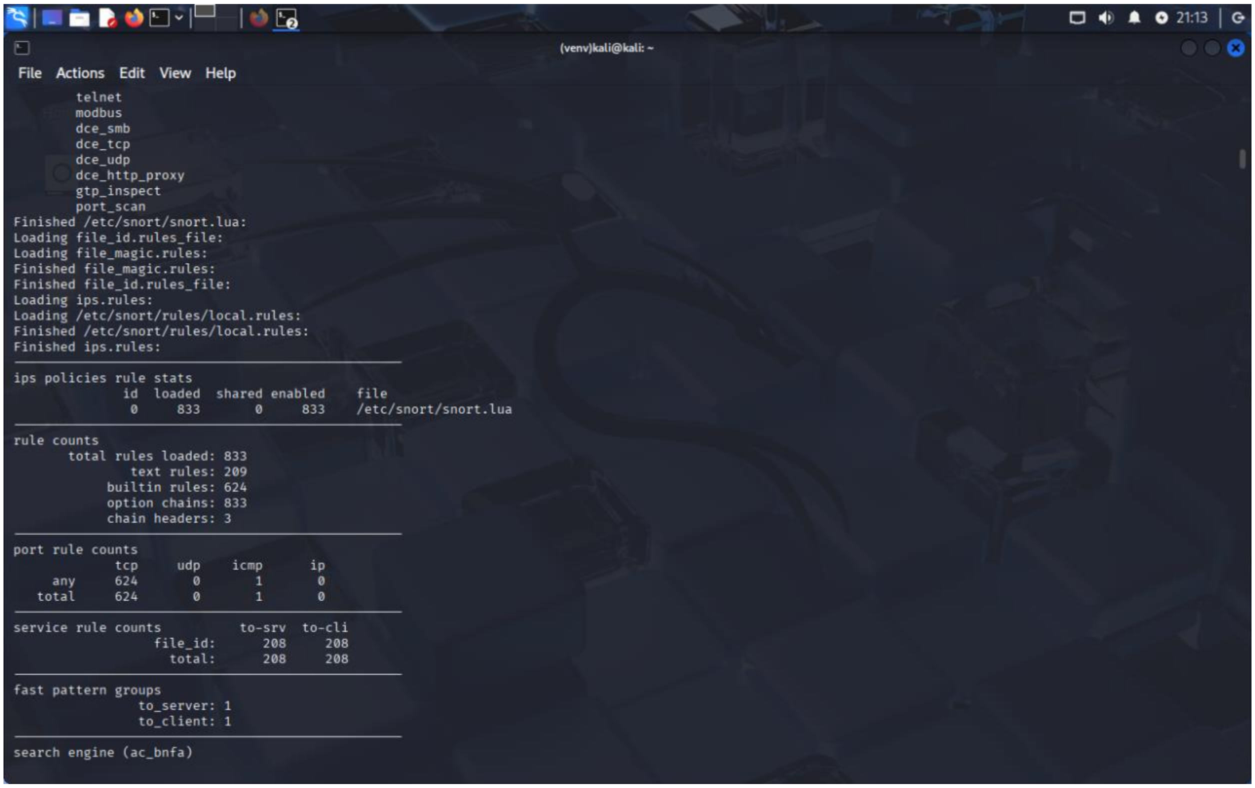

21:12
(venv)kali@kali: ~
File Actions Edit View Help
┌──(venv)─(kali㉿kali)-[~]
└─$ snort -V
-*> Snort++ <*-
Version 3.1.82.0
By Martin Roesch & The Snort Team
http://snort.org/contact#team
Copyright (C) 2014-2024 Cisco and/or its affiliates. All rights reserve
d.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using DAQ version 3.0.12
Using LuaJIT version 2.1.1700206165
Using OpenSSL 3.5.0 8 Apr 2025
Using libpcap version 1.10.5 (with TPACKET_V3)
Using PCRE version 8.39 2016-06-14
Using ZLIB version 1.3.1
Using LZMA version 5.8.1
┌──(venv)─(kali㉿kali)-[~]
└─$ sudo apt update
sudo apt install snort -y
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main arm64 Packages [20.8 MB]
Get:3 http://kali.download/kali kali-rolling/main arm64 Contents (deb) [50.5 MB]
Get:4 http://kali.download/kali kali-rolling/contrib arm64 Packages [103 kB]
Get:5 http://kali.download/kali kali-rolling/contrib arm64 Contents (deb) [246 kB]
Fetched 71.7 MB in 7s (9,637 kB/s)
361 packages can be upgraded. Run 'apt list --upgradable' to see them.
snort is already the newest version (3.1.82.0-0kali1+b1).
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 361
┌──(venv)─(kali㉿kali)-[~]
└─$ sudo mkdir -p /var/log/snort
┌──(venv)─(kali㉿kali)-[~]
└─$ ls -ld /var/log/snort

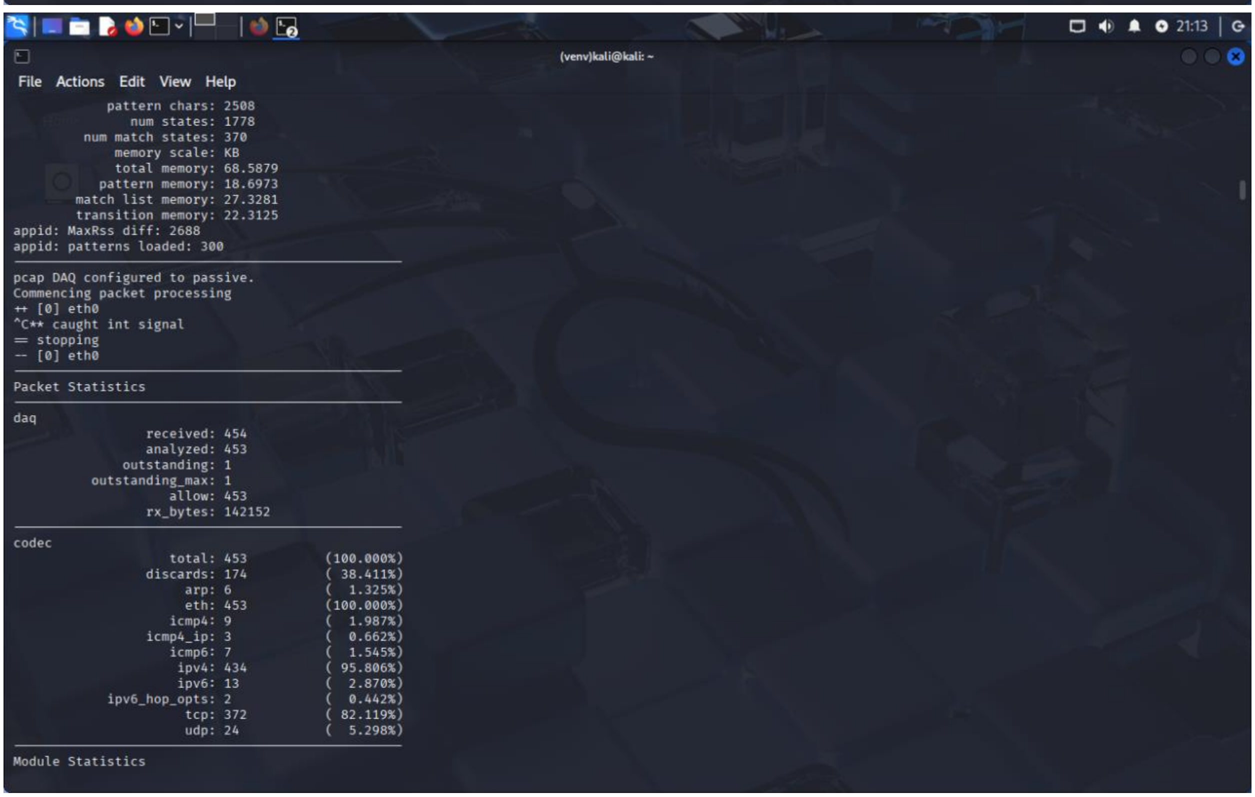


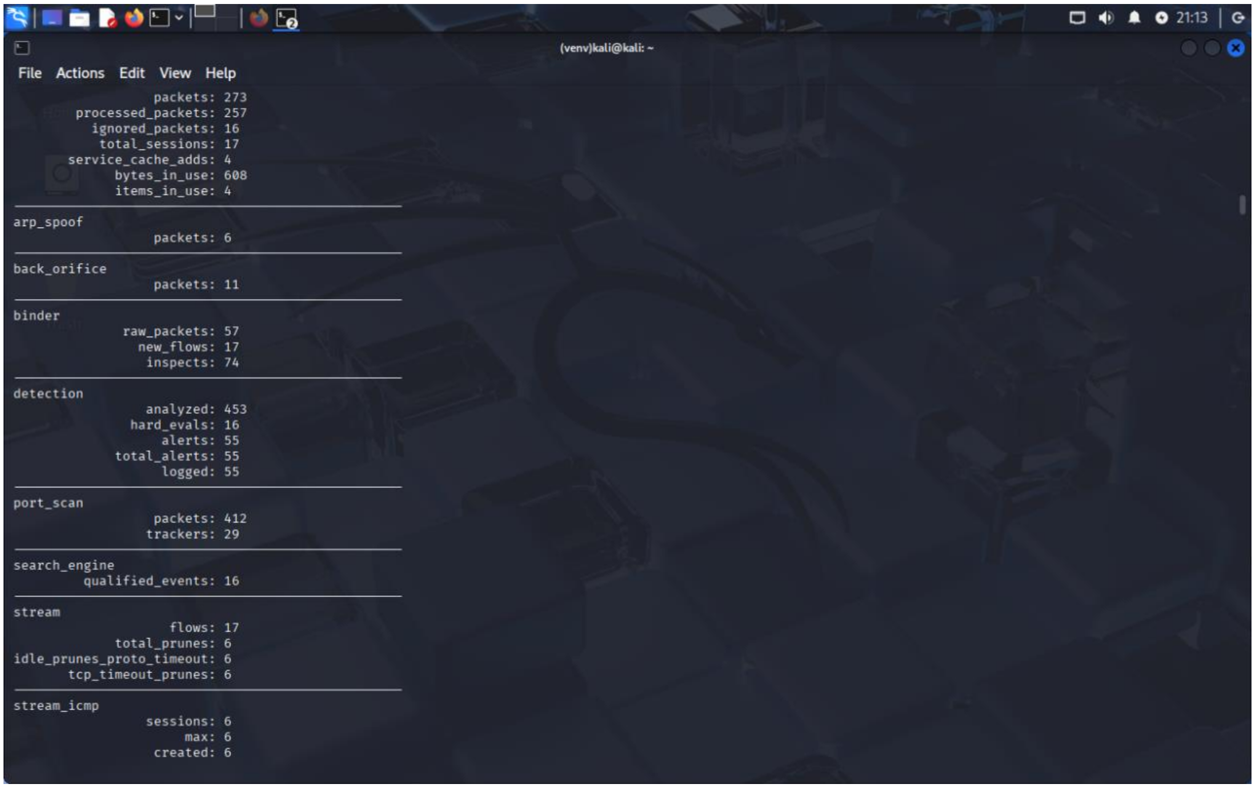
File Actions Edit View Help
(venv)kali@kali: ~
┌──(venv)─(kali㉿kali)-[~]
└─$ ls -ld /var/log/snort
drwxr-sr-x 2 snort adm 4096 Jul 13 19:38 /var/log/snort
┌──(venv)─(kali㉿kali)-[~]
└─$ echo 'alert icmp any any → any any (msg:"ICMP test detected"; sid:1000001; rev:1;)' | sudo tee /etc/snort/rules/local.rules
alert icmp any any → any any (msg:"ICMP test detected"; sid:1000001; rev:1;)
┌──(venv)─(kali㉿kali)-[~]
└─$ cat /etc/snort/rules/local.rules
alert icmp any any → any any (msg:"ICMP test detected"; sid:1000001; rev:1;)
┌──(venv)─(kali㉿kali)-[~]
└─$ sudo nano /etc/snort/snort.lua
┌──(venv)─(kali㉿kali)-[~]
└─$ sudo snort -i eth0 -c /etc/snort/snort.lua -l /var/log/snort
o")~ Snort++ 3.1.82.0
Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
smtp
ftp_server
ftp_client
ftp_data
http_inspect
http2_inspect
file_policy
js_norm
appid
wizard
binder
active
alerts

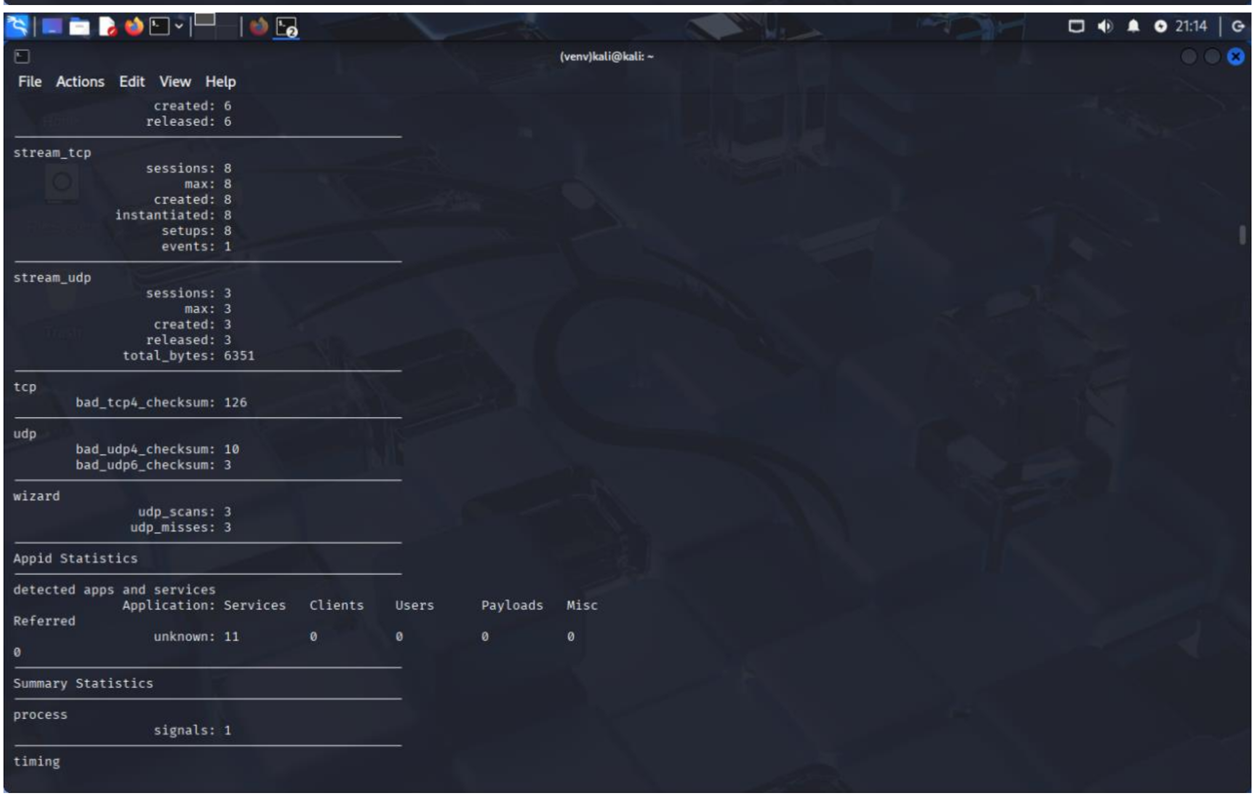


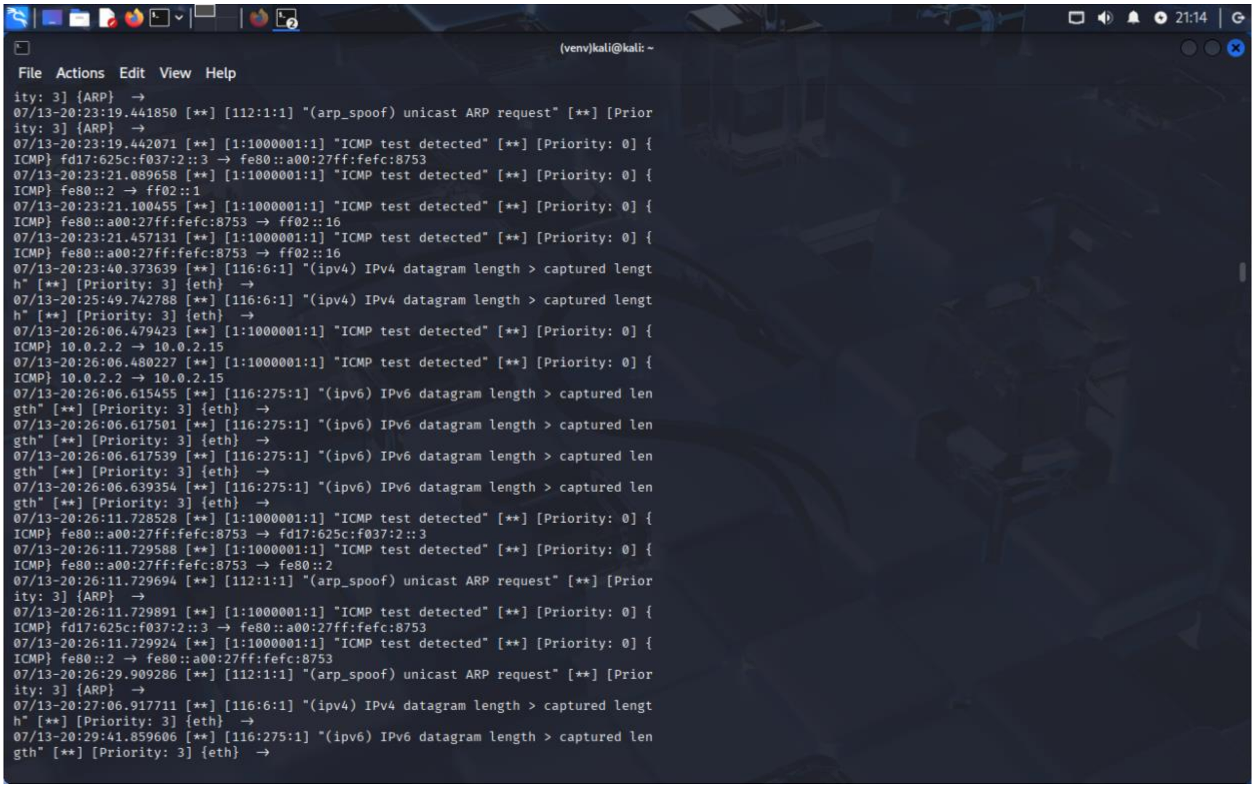
File Actions Edit View Help
(venv)kali@kali: ~
alerts
daq
decode
host_cache
host_tracker
network
packets
process
search_engine
so_proxy
stream
stream_ip
stream_icmp
stream_tcp
stream_udp
stream_user
alert_fast
ips
classifications
references
file_id
dce_http_server
trace
s7commplus
mms
iec104
dnp3
cip
hosts
ssl
ssh
sip
rpc_decode
pop
normalizer
stream_file
arp_spoof
back_orifice
dns
imap
netflow
output
telnet


(venv)kali@kali: ~
File Actions Edit View Help
telnet
modbus
dce_smb
dce_tcp
dce_udp
dce_http_proxy
gtp_inspect
port_scan
Finished /etc/snort/snort.lua:
Loading file_id.rules_file:
Loading file_magic.rules:
Finished file_magic.rules:
Finished file_id.rules_file:
Loading ips.rules:
Loading /etc/snort/rules/local.rules:
Finished /etc/snort/rules/local.rules:
Finished ips.rules:
ips policies rule stats
id loaded shared enabled file
0 833 0 833 /etc/snort/snort.lua
rule counts
total rules loaded: 833
text rules: 209
builtin rules: 624
option chains: 833
chain headers: 3
port rule counts
tcp udp icmp ip
any 624 0 1 0
total 624 0 1 0
service rule counts to-srv to-cli
file_id: 208 208
total: 208 208
fast pattern groups
to_server: 1
to_client: 1
search engine (ac_bnfa)

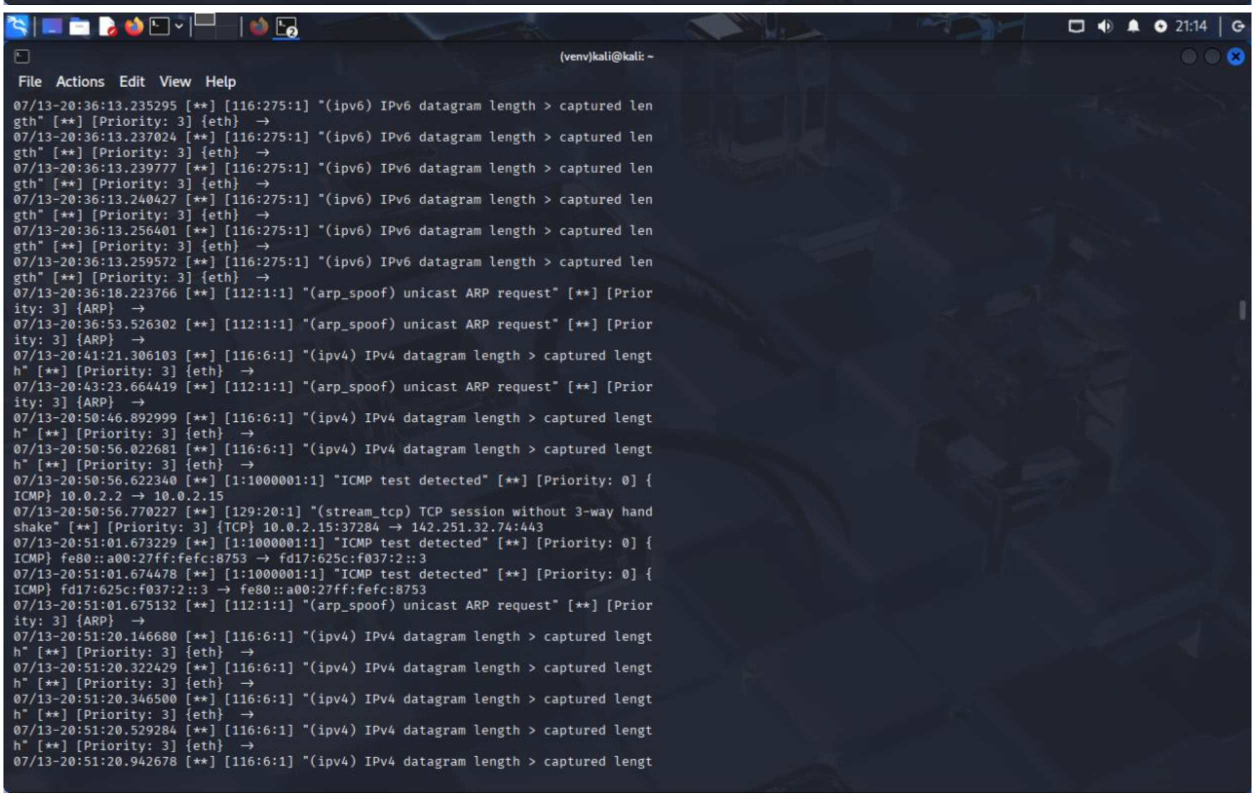

(venv)kali@kali: ~
File Actions Edit View Help
pattern chars: 2508
num states: 1778
num match states: 370
memory scale: KB
total memory: 68.5879
pattern memory: 18.6973
match list memory: 27.3281
transition memory: 22.3125
appid: MaxRss diff: 2688
appid: patterns loaded: 300
pcap DAQ configured to passive.
Commencing packet processing
++ [0] eth0
^C** caught int signal
== stopping
-- [0] eth0
Packet Statistics
daq
received: 454
analyzed: 453
outstanding: 1
outstanding_max: 1
allow: 453
rx_bytes: 142152
codec
total: 453 (100.000%)
discards: 174 (38.411%)
arp: 6 (1.325%)
eth: 453 (100.000%)
icmp4: 9 (1.987%)
icmp4_ip: 3 (0.662%)
icmp6: 7 (1.545%)
ipv4: 434 (95.806%)
ipv6: 13 (2.870%)
ipv6_hop_opts: 2 (0.442%)
tcp: 372 (82.119%)
udp: 24 (5.298%)
Module Statistics


21:13
(venv)kali@kali: ~
File Actions Edit View Help
packets: 273
processed_packets: 257
ignored_packets: 16
total_sessions: 17
service_cache_adds: 4
bytes_in_use: 608
items_in_use: 4
arp_spoof
packets: 6
back_orifice
packets: 11
binder
raw_packets: 57
new_flows: 17
inspects: 74
detection
analyzed: 453
hard_evals: 16
alerts: 55
total_alerts: 55
logged: 55
port_scan
packets: 412
trackers: 29
search_engine
qualified_events: 16
stream
flows: 17
total_prunes: 6
idle_prunes_proto_timeout: 6
tcp_timeout_prunes: 6
stream_icmp
sessions: 6
max: 6
created: 6

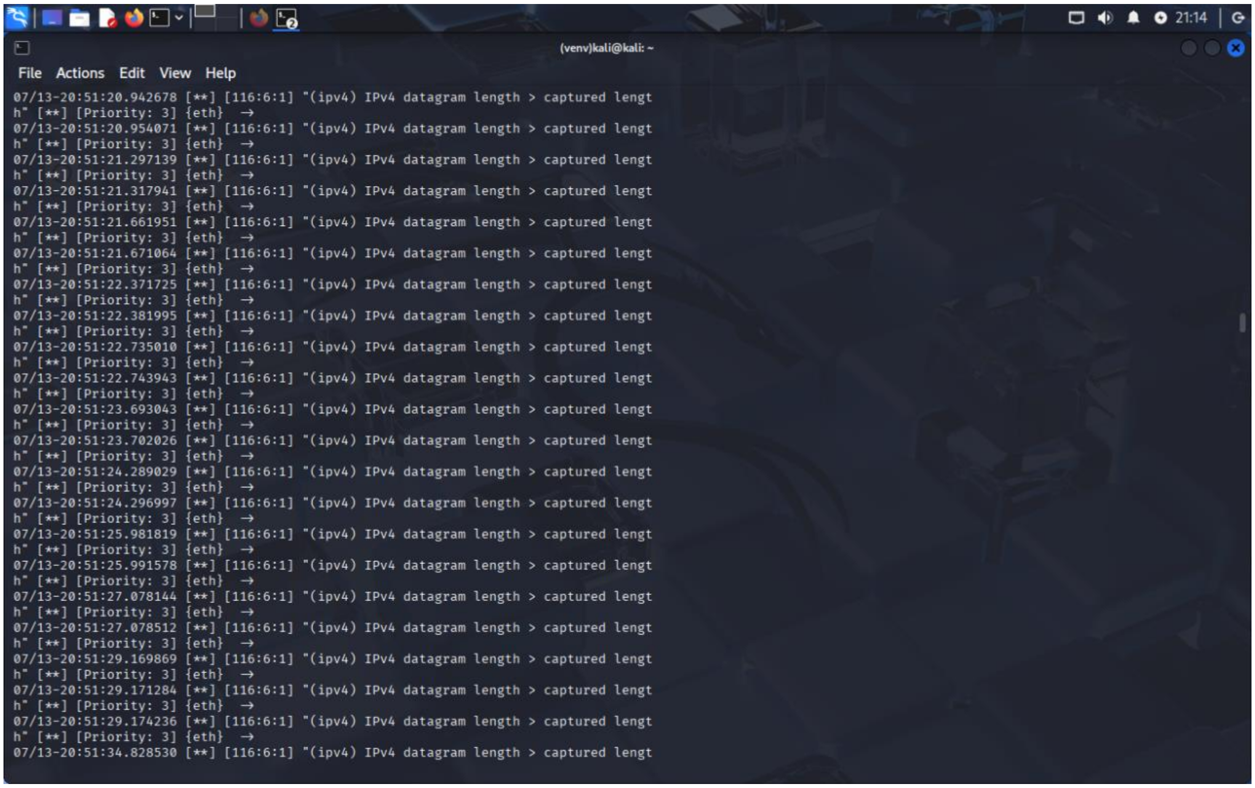

21:14
(venv)kali@kali: ~
File Actions Edit View Help
created: 6
released: 6
stream_tcp
sessions: 8
max: 8
created: 8
instantiated: 8
setups: 8
events: 1
stream_udp
sessions: 3
max: 3
created: 3
released: 3
total_bytes: 6351
tcp
bad_tcp4_checksum: 126
udp
bad_udp4_checksum: 10
bad_udp6_checksum: 3
wizard
udp_scans: 3
udp_misses: 3
Appid Statistics
detected apps and services
Application: Services Clients Users Payloads Misc
Referred
unknown: 11 0 0 0 0
0
Summary Statistics
process
signals: 1
timing



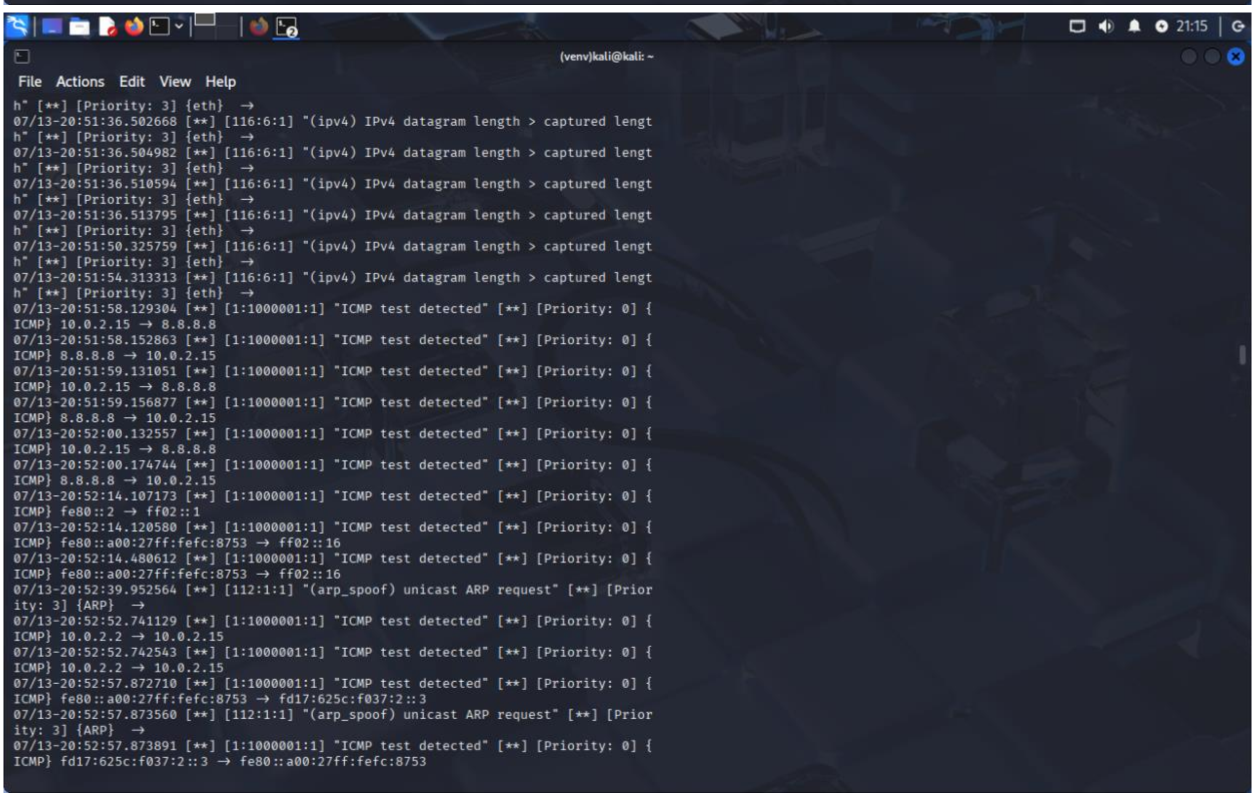
```
File  Actions  Edit  View  Help
ity: 3] {ARP}  →
07/13-20:23:19.441850 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:23:19.442071 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
07/13-20:23:21.089658 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::2 → ff02::1
07/13-20:23:21.100455 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → ff02::16
07/13-20:23:21.457131 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → ff02::16
07/13-20:23:40.373639 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:25:49.742788 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:26:06.479423 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:26:06.480227 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:26:06.615455 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:26:06.617501 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:26:06.617539 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:26:06.639354 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:26:11.728528 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-20:26:11.729588 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fe80::2
07/13-20:26:11.729694 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:26:11.729891 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
07/13-20:26:11.729924 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::2 → fe80::a00:27ff:fefc:8753
07/13-20:26:29.909286 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:27:06.917711 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:29:41.859606 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
```



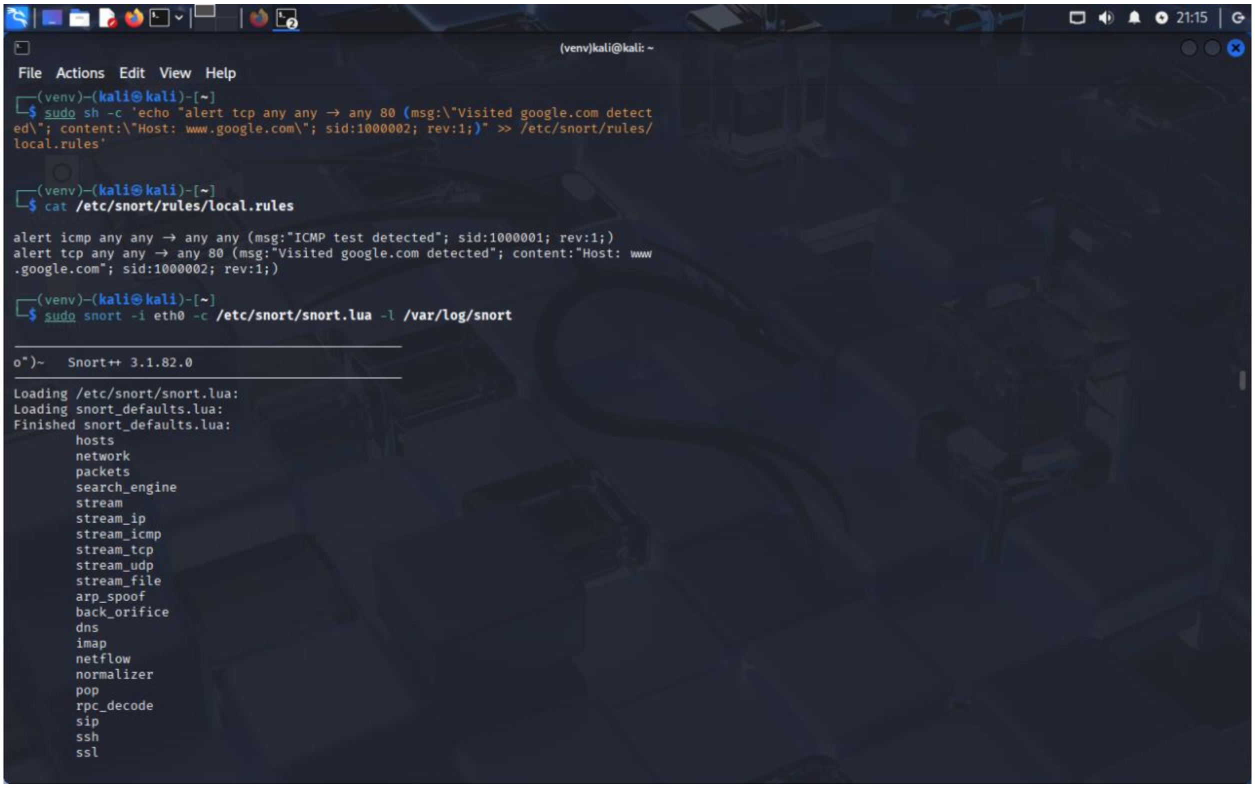
```
File  Actions  Edit  View  Help
07/13-20:36:13.235295 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:36:13.237024 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:36:13.239777 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:36:13.240427 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:36:13.256401 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:36:13.259572 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:36:18.223766 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:36:53.526302 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:41:21.306103 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:43:23.664419 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:50:46.892999 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:50:56.022681 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:50:56.622340 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:50:56.770227 [**] [129:20:1] "(stream_tcp) TCP session without 3-way hand
shake" [**] [Priority: 3] {TCP} 10.0.2.15:37284 → 142.251.32.74:443
07/13-20:51:01.673229 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-20:51:01.674478 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
07/13-20:51:01.675132 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:51:20.146680 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:51:20.322429 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:51:20.346500 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:51:20.529284 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:51:20.942678 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
```

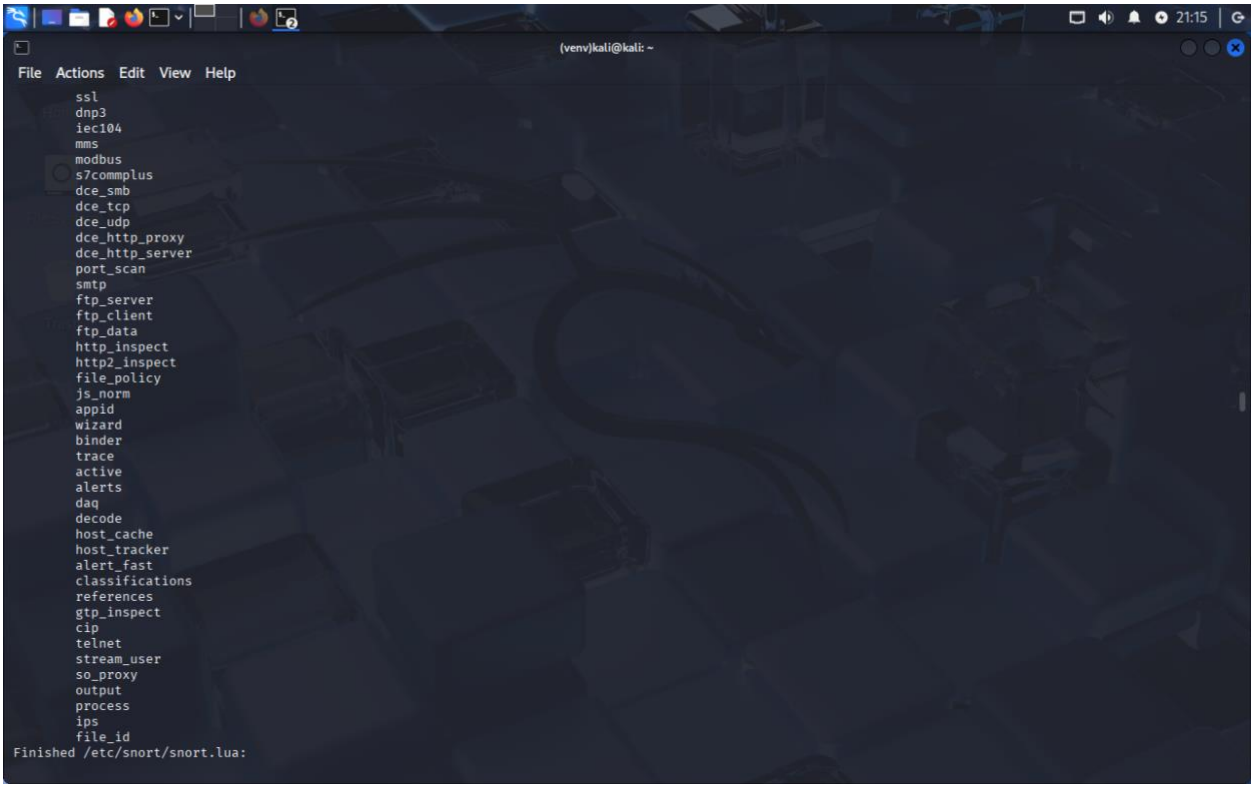


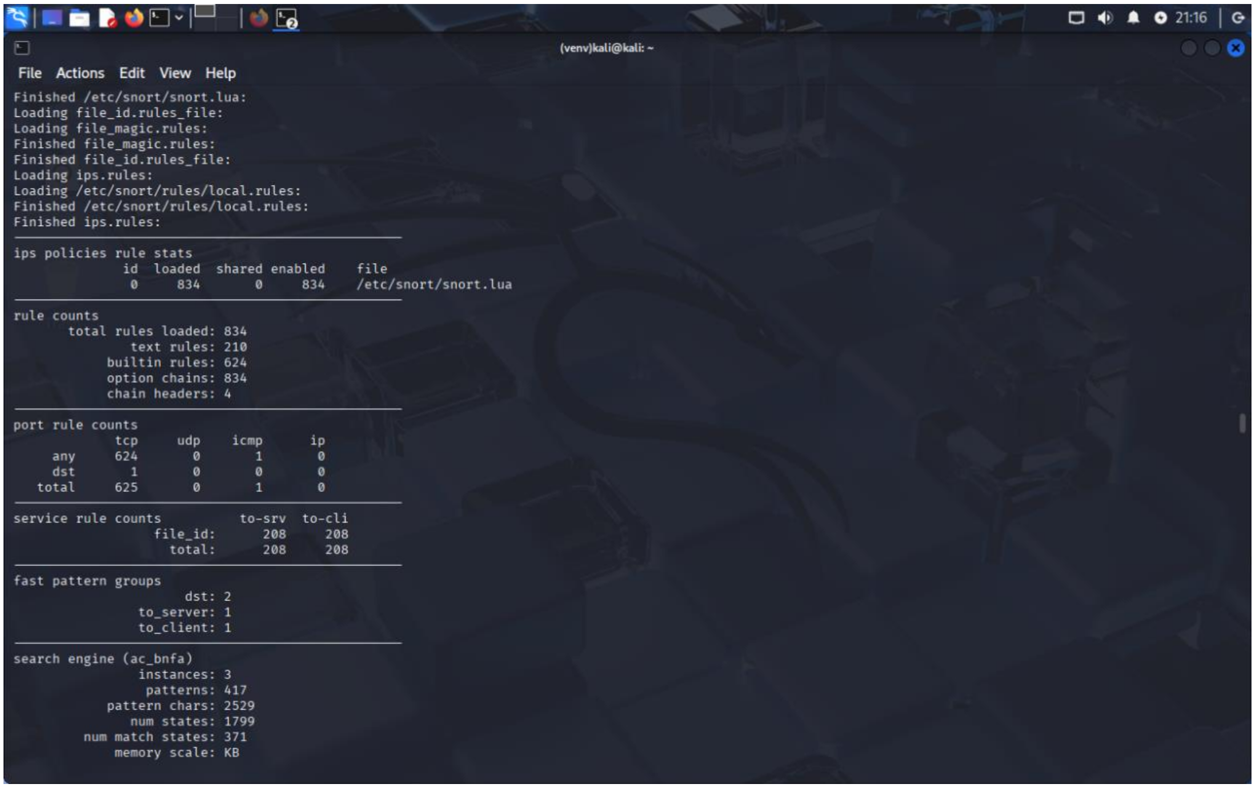
```
(venv)kali@kali: ~
File Actions Edit View Help
07/13-20:51:20.942678 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:20.954071 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:21.297139 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:21.317941 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:21.661951 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:21.671064 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:22.371725 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:22.381995 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:22.735010 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:22.743943 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:23.693043 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:23.702026 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:24.289029 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:24.296997 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:25.981819 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:25.991578 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:27.078144 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:27.078512 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:29.169869 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:29.171284 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:29.174236 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:34.828530 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
```



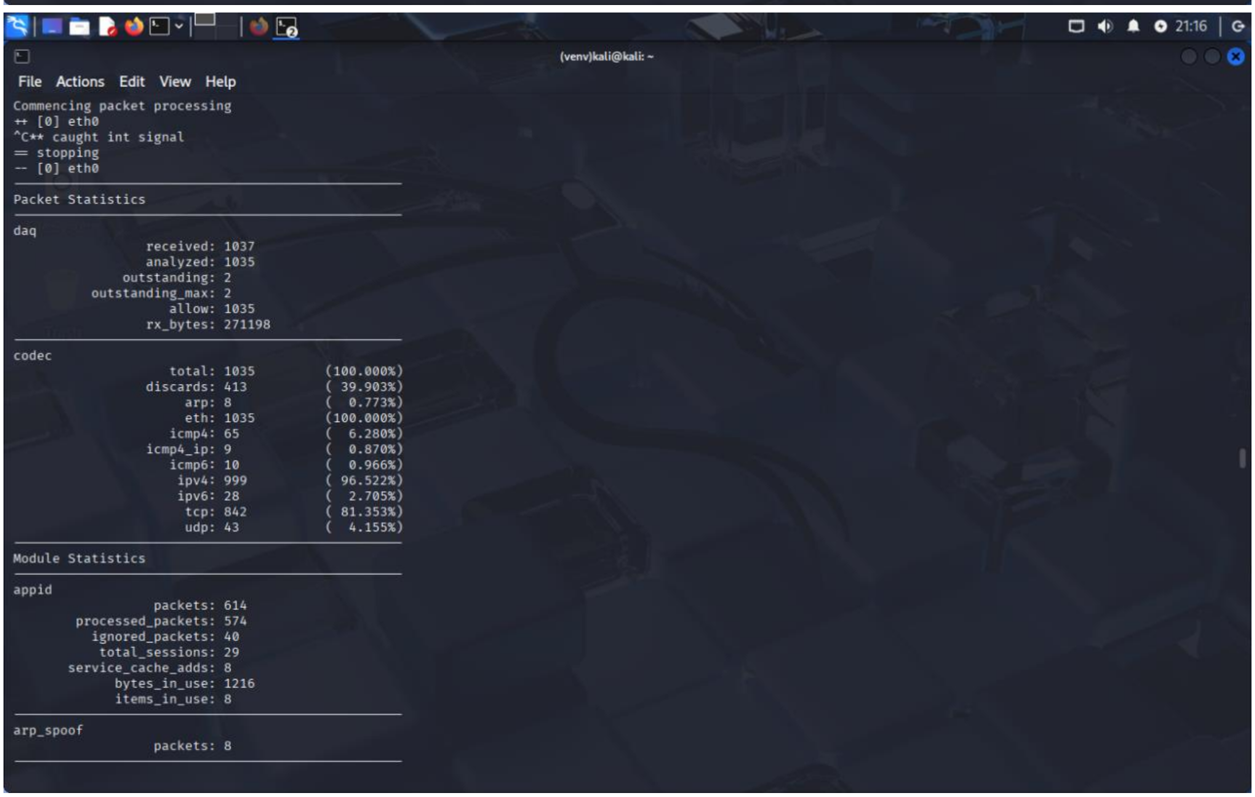
```
(venv)kali@kali: ~
File Actions Edit View Help
h" [**] [Priority: 3] {eth} →
07/13-20:51:36.502668 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:36.504982 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:36.510594 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:36.513795 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:50.325759 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:54.313313 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:58.129304 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 8.8.8.8
07/13-20:51:58.152863 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 8.8.8.8 → 10.0.2.15
07/13-20:51:59.131051 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 8.8.8.8
07/13-20:51:59.156877 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 8.8.8.8 → 10.0.2.15
07/13-20:52:00.132557 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 8.8.8.8
07/13-20:52:00.174744 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 8.8.8.8 → 10.0.2.15
07/13-20:52:14.107173 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::2 → ff02::1
07/13-20:52:14.120580 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → ff02::16
07/13-20:52:14.480612 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → ff02::16
07/13-20:52:39.952564 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP} →
07/13-20:52:52.741129 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:52:52.742543 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:52:57.872710 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-20:52:57.873560 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP} →
07/13-20:52:57.873891 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
```


21:15
(venv)kali@kali: ~
File Actions Edit View Help
┌──(venv)─(kali㉿kali)-[~]
└─$ sudo sh -c 'echo "alert tcp any any → any 80 (msg:\"Visited google.com detected\"; content:\"Host: www.google.com\"; sid:1000002; rev:1;)" >> /etc/snort/rules/local.rules'
┌──(venv)─(kali㉿kali)-[~]
└─$ cat /etc/snort/rules/local.rules
alert icmp any any → any any (msg:"ICMP test detected"; sid:1000001; rev:1;)
alert tcp any any → any 80 (msg:"Visited google.com detected"; content:"Host: www.google.com"; sid:1000002; rev:1;)
┌──(venv)─(kali㉿kali)-[~]
└─$ sudo snort -i eth0 -c /etc/snort/snort.lua -l /var/log/snort
o")~ Snort++ 3.1.82.0
Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
hosts
network
packets
search_engine
stream
stream_ip
stream_icmp
stream_tcp
stream_udp
stream_file
arp_spoof
back_orifice
dns
imap
netflow
normalizer
pop
rpc_decode
sip
ssh
ssl

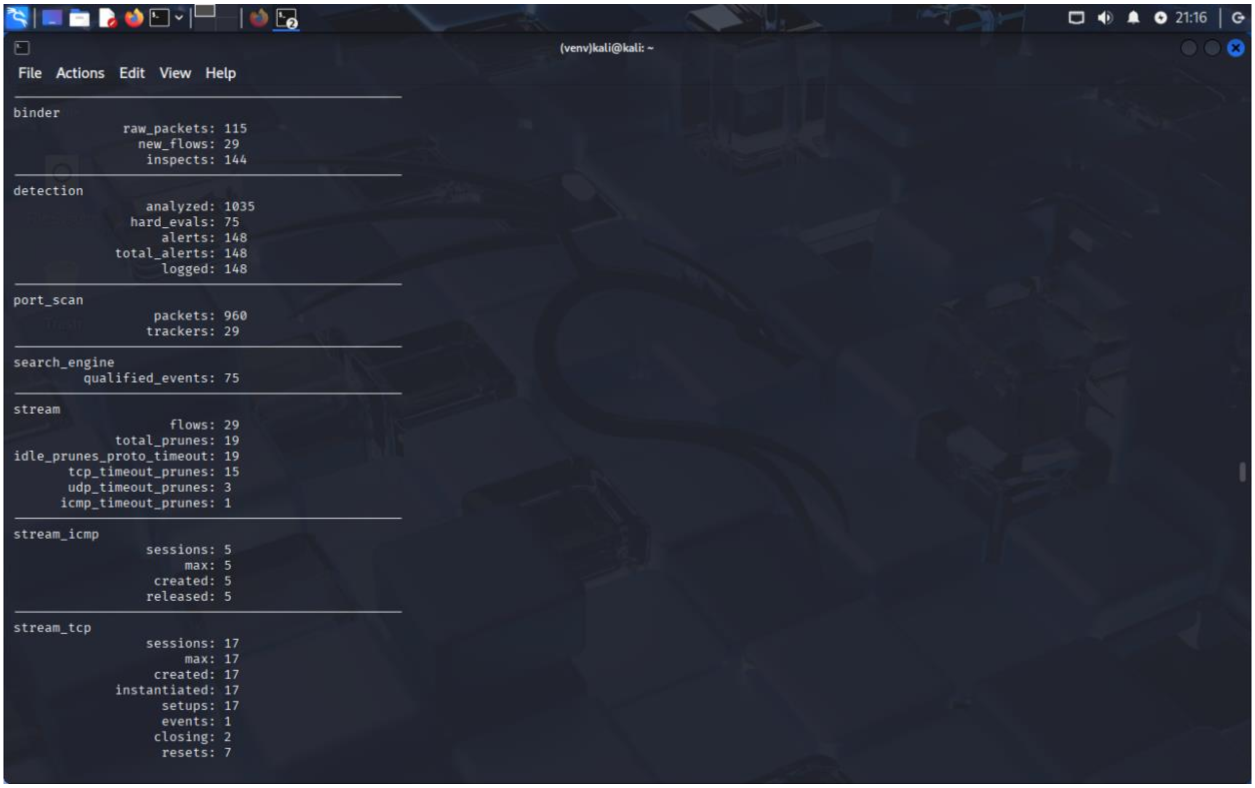

21:15
(venv)kali@kali: ~
File Actions Edit View Help
ssl
dnp3
iec104
mms
modbus
s7commplus
dce_smb
dce_tcp
dce_udp
dce_http_proxy
dce_http_server
port_scan
smtp
ftp_server
ftp_client
ftp_data
http_inspect
http2_inspect
file_policy
js_norm
appid
wizard
binder
trace
active
alerts
daq
decode
host_cache
host_tracker
alert_fast
classifications
references
gtp_inspect
cip
telnet
stream_user
so_proxy
output
process
ips
file_id
Finished /etc/snort/snort.lua:

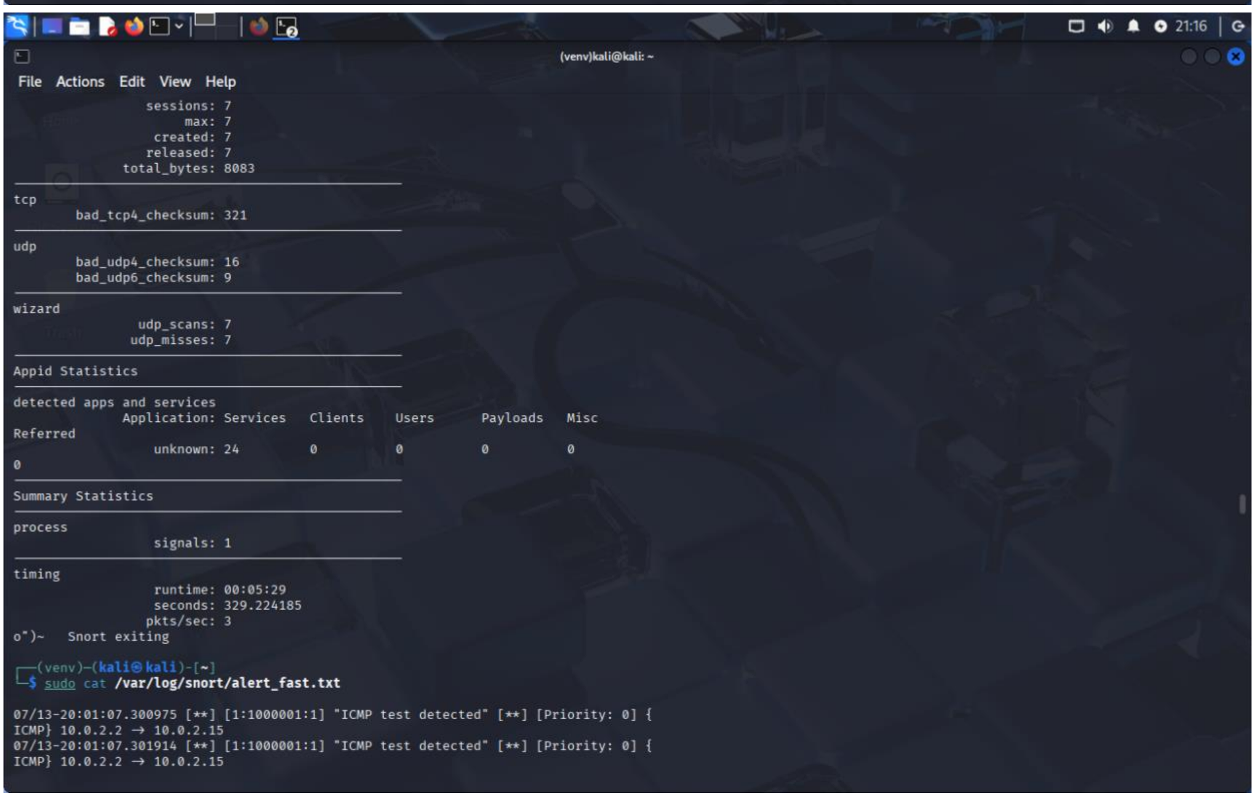


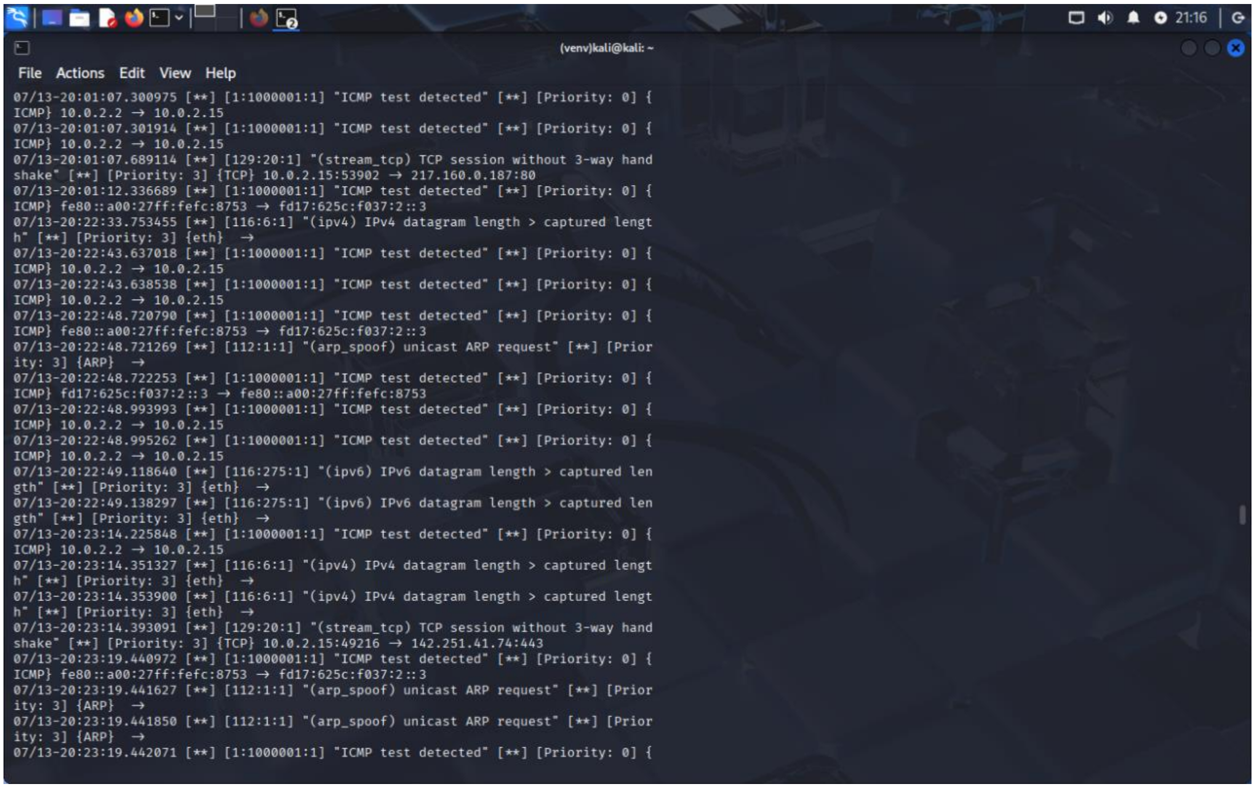

```
Finished /etc/snort/snort.lua:
Loading file_id.rules_file:
Loading file_magic.rules:
Finished file_magic.rules:
Finished file_id.rules_file:
Loading ips.rules:
Loading /etc/snort/rules/local.rules:
Finished /etc/snort/rules/local.rules:
Finished ips.rules:
--------------------------------------------------
ips policies rule stats
              id  loaded  shared enabled    file
               0     834       0     834    /etc/snort/snort.lua
--------------------------------------------------
rule counts
       total rules loaded: 834
               text rules: 210
            builtin rules: 624
            option chains: 834
            chain headers: 4
--------------------------------------------------
port rule counts
             tcp     udp    icmp      ip
     any     624       0       1       0
     dst       1       0       0       0
   total     625       0       1       0
--------------------------------------------------
service rule counts          to-srv  to-cli
                  file_id:      208     208
                    total:      208     208
--------------------------------------------------
fast pattern groups
                      dst: 2
                to_server: 1
                to_client: 1
--------------------------------------------------
search engine (ac_bnfa)
                instances: 3
                 patterns: 417
            pattern chars: 2529
               num states: 1799
         num match states: 371
             memory scale: KB
```



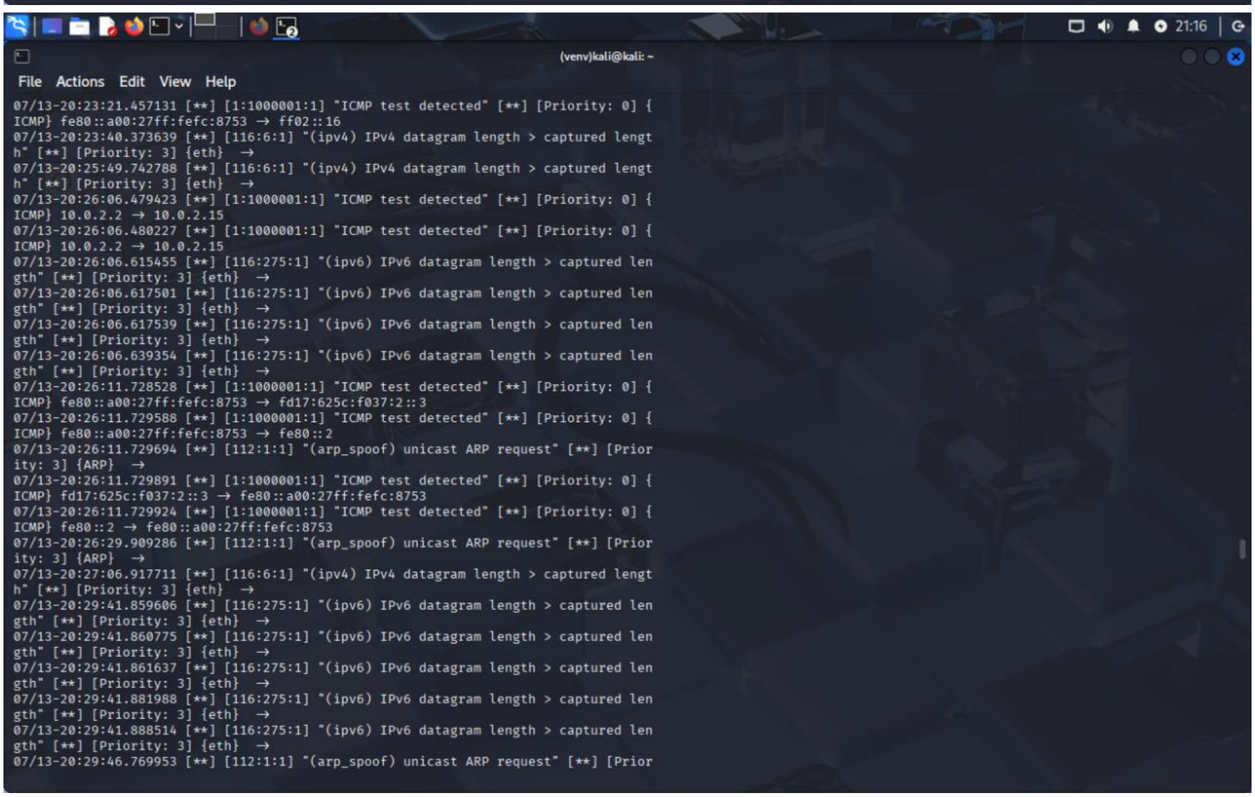

```
Commencing packet processing
++ [0] eth0
^C** caught int signal
== stopping
-- [0] eth0
--------------------------------------------------
Packet Statistics
--------------------------------------------------
daq
                 received: 1037
                 analyzed: 1035
              outstanding: 2
          outstanding_max: 2
                    allow: 1035
                 rx_bytes: 271198
--------------------------------------------------
codec
                    total: 1035         (100.000%)
                 discards: 413          ( 39.903%)
                      arp: 8            (  0.773%)
                      eth: 1035         (100.000%)
                    icmp4: 65           (  6.280%)
                 icmp4_ip: 9            (  0.870%)
                    icmp6: 10           (  0.966%)
                     ipv4: 999          ( 96.522%)
                     ipv6: 28           (  2.705%)
                      tcp: 842          ( 81.353%)
                      udp: 43           (  4.155%)
--------------------------------------------------
Module Statistics
--------------------------------------------------
appid
                  packets: 614
        processed_packets: 574
          ignored_packets: 40
           total_sessions: 29
       service_cache_adds: 8
             bytes_in_use: 1216
             items_in_use: 8
--------------------------------------------------
arp_spoof
                  packets: 8
--------------------------------------------------
```


(venv)kali@kali: ~
File Actions Edit View Help
binder
raw_packets: 115
new_flows: 29
inspects: 144
detection
analyzed: 1035
hard_evals: 75
alerts: 148
total_alerts: 148
logged: 148
port_scan
packets: 960
trackers: 29
search_engine
qualified_events: 75
stream
flows: 29
total_prunes: 19
idle_prunes_proto_timeout: 19
tcp_timeout_prunes: 15
udp_timeout_prunes: 3
icmp_timeout_prunes: 1
stream_icmp
sessions: 5
max: 5
created: 5
released: 5
stream_tcp
sessions: 17
max: 17
created: 17
instantiated: 17
setups: 17
events: 1
closing: 2
resets: 7

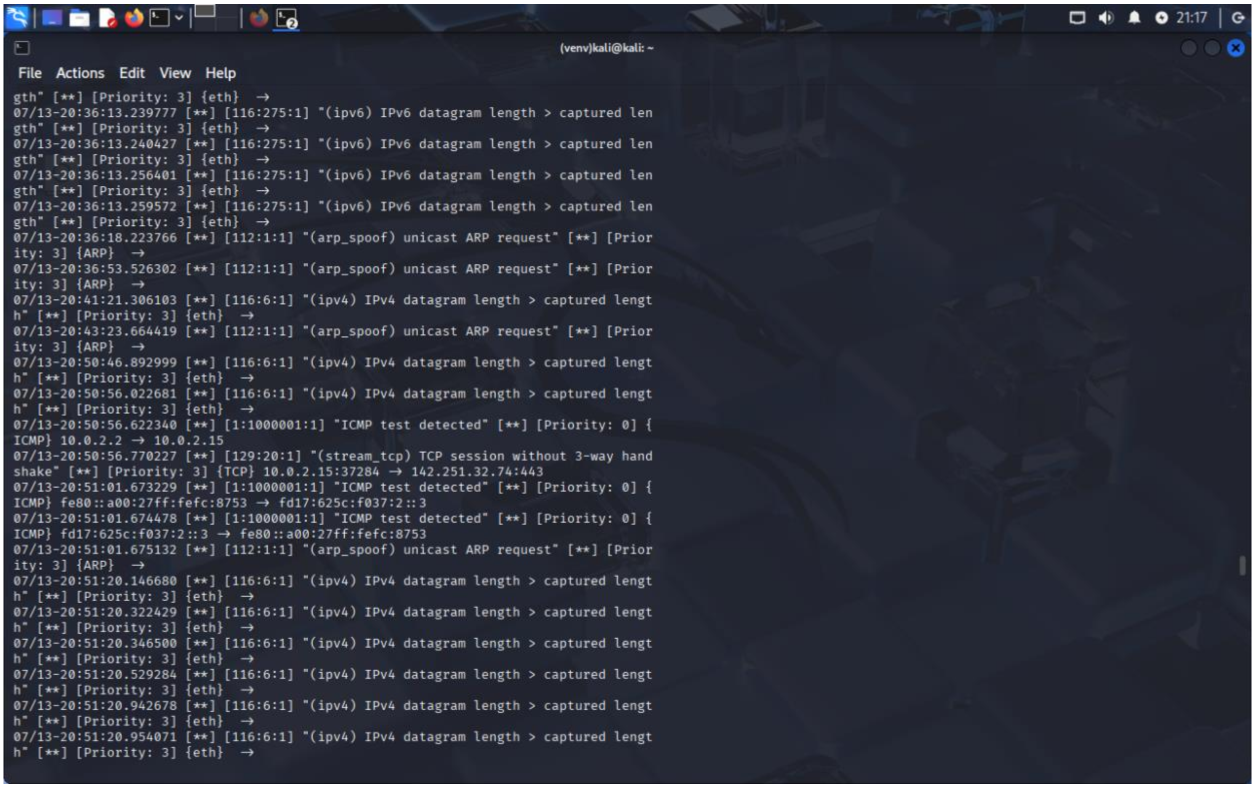

(venv)kali@kali: ~
File Actions Edit View Help
sessions: 7
max: 7
created: 7
released: 7
total_bytes: 8083
tcp
bad_tcp4_checksum: 321
udp
bad_udp4_checksum: 16
bad_udp6_checksum: 9
wizard
udp_scans: 7
udp_misses: 7
Appid Statistics
detected apps and services
Application: Services Clients Users Payloads Misc
Referred
unknown: 24 0 0 0 0
0
Summary Statistics
process
signals: 1
timing
runtime: 00:05:29
seconds: 329.224185
pkts/sec: 3
o")~ Snort exiting
(venv)-(kali㉿kali)-[~]
$ sudo cat /var/log/snort/alert_fast.txt
07/13-20:01:07.300975 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 10.0.2.2 -> 10.0.2.15
07/13-20:01:07.301914 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 10.0.2.2 -> 10.0.2.15



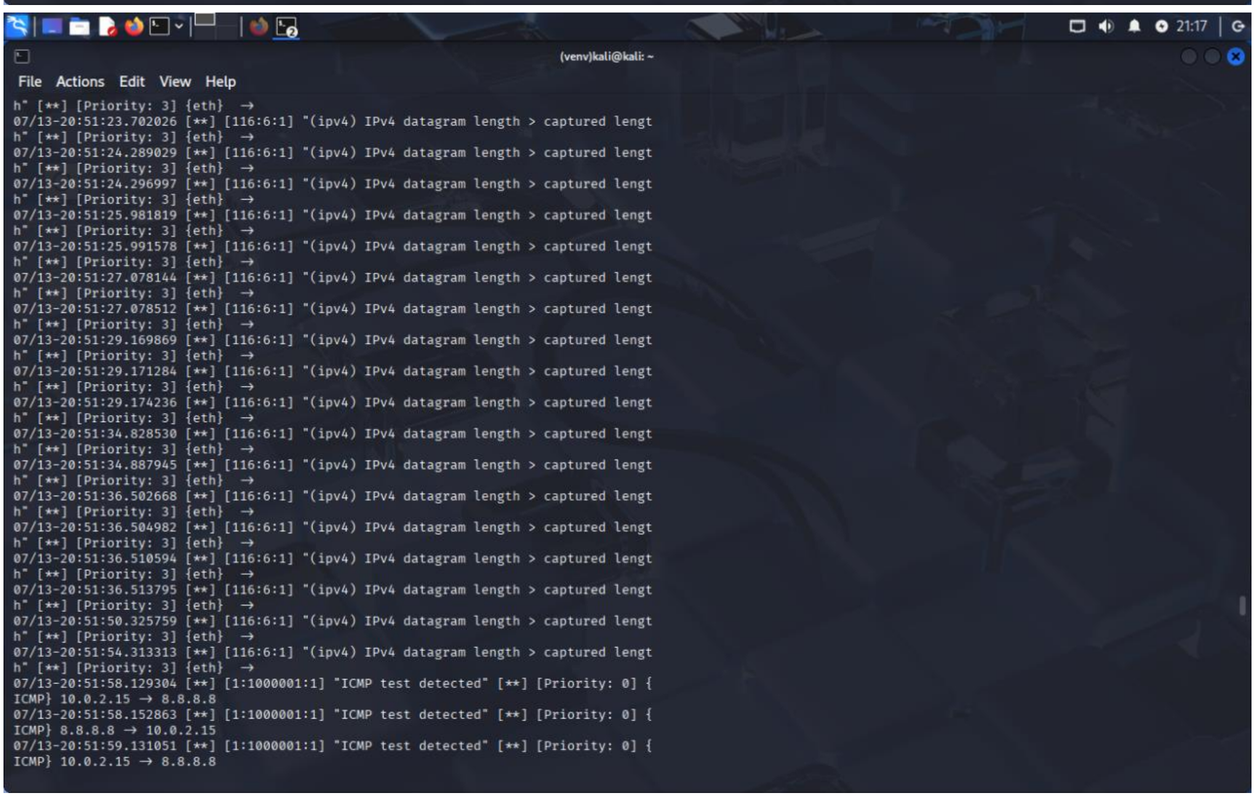

```
07/13-20:01:07.300975 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:01:07.301914 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:01:07.689114 [**] [129:20:1] "(stream_tcp) TCP session without 3-way hand
shake" [**] [Priority: 3] {TCP} 10.0.2.15:53902 → 217.160.0.187:80
07/13-20:01:12.336689 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-20:22:33.753455 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:22:43.637018 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:22:43.638538 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:22:48.720790 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-20:22:48.721269 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:22:48.722253 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
07/13-20:22:48.993993 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:22:48.995262 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:22:49.118640 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:22:49.138297 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:23:14.225848 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:23:14.351327 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:23:14.353900 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:23:14.393091 [**] [129:20:1] "(stream_tcp) TCP session without 3-way hand
shake" [**] [Priority: 3] {TCP} 10.0.2.15:49216 → 142.251.41.74:443
07/13-20:23:19.440972 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-20:23:19.441627 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:23:19.441850 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:23:19.442071 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
```



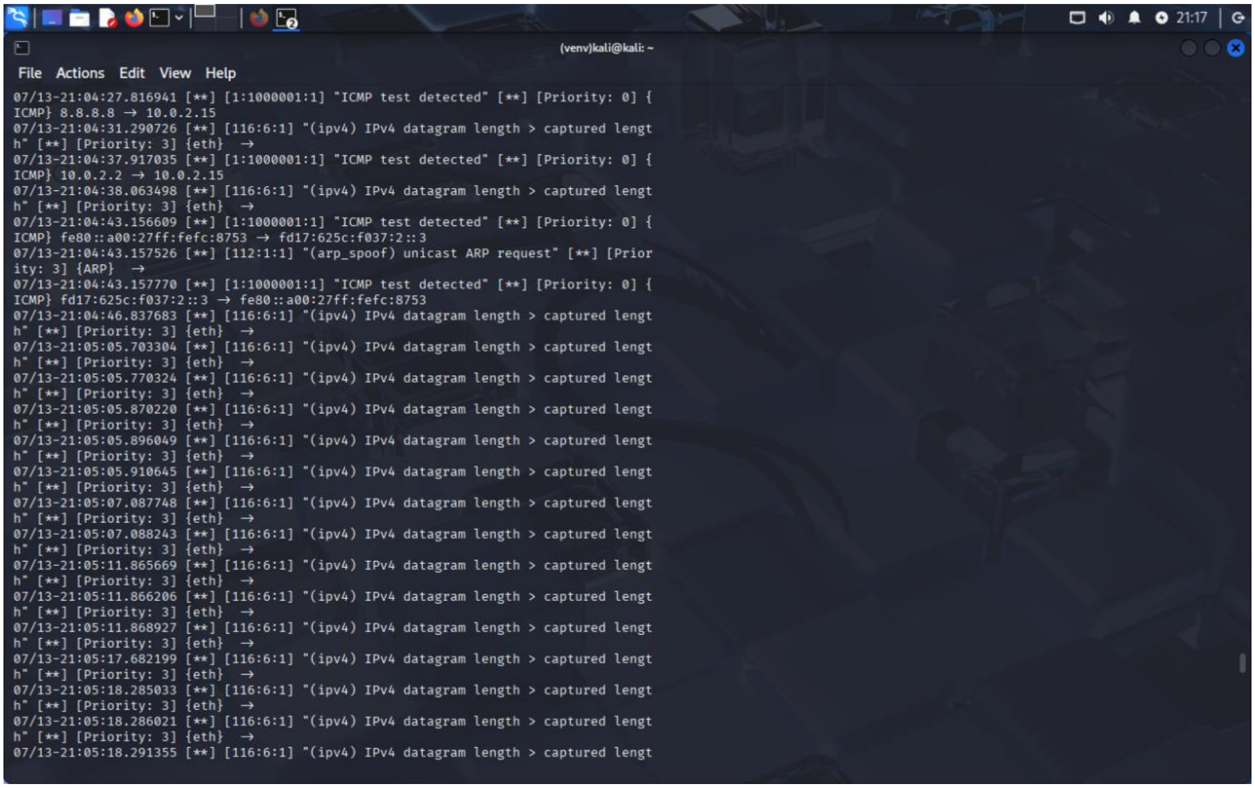

```
07/13-20:23:21.457131 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → ff02::16
07/13-20:23:40.373639 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:25:49.742788 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:26:06.479423 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:26:06.480227 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:26:06.615455 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:26:06.617501 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:26:06.617539 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:26:06.639354 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:26:11.728528 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-20:26:11.729588 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fe80::2
07/13-20:26:11.729694 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:26:11.729891 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
07/13-20:26:11.729924 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::2 → fe80::a00:27ff:fefc:8753
07/13-20:26:29.909286 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP}  →
07/13-20:27:06.917711 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth}  →
07/13-20:29:41.859606 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:29:41.860775 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:29:41.861637 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:29:41.881988 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:29:41.888514 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth}  →
07/13-20:29:46.769953 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
```

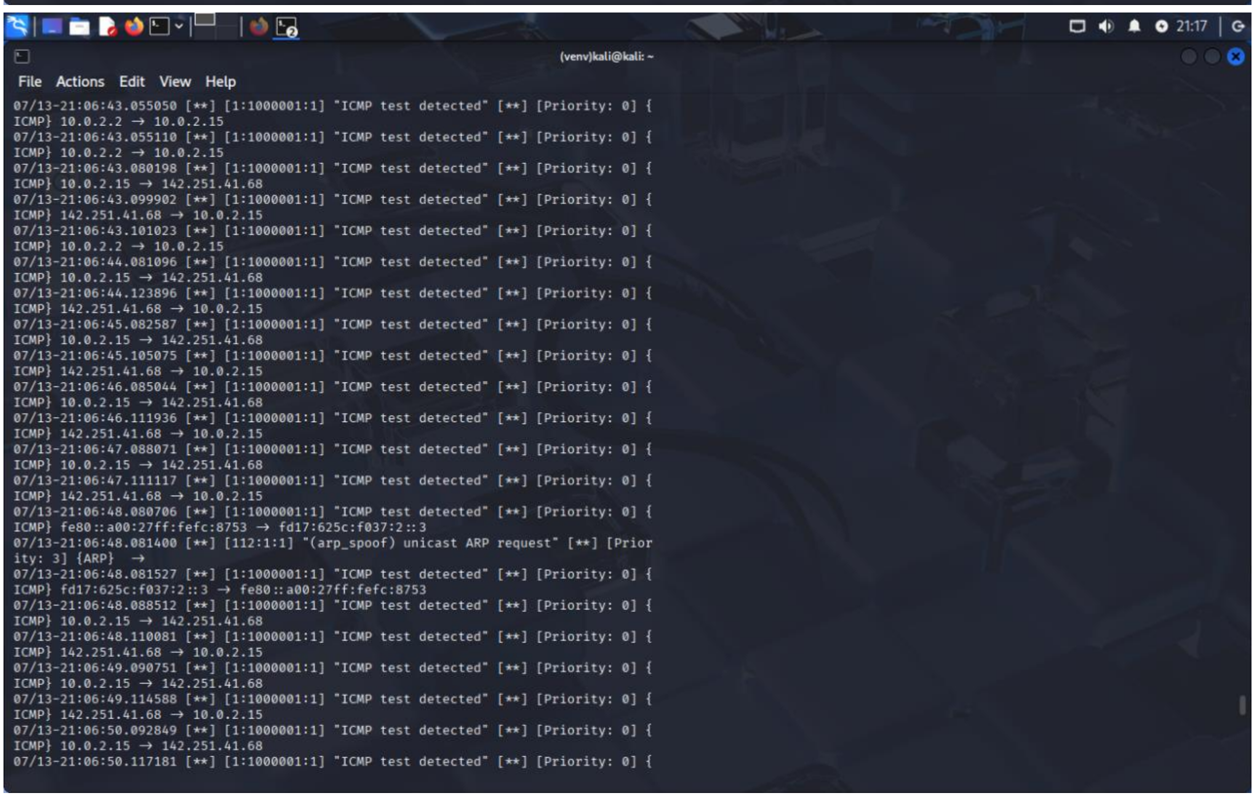

21:17
(venv)kali@kali: ~
File Actions Edit View Help

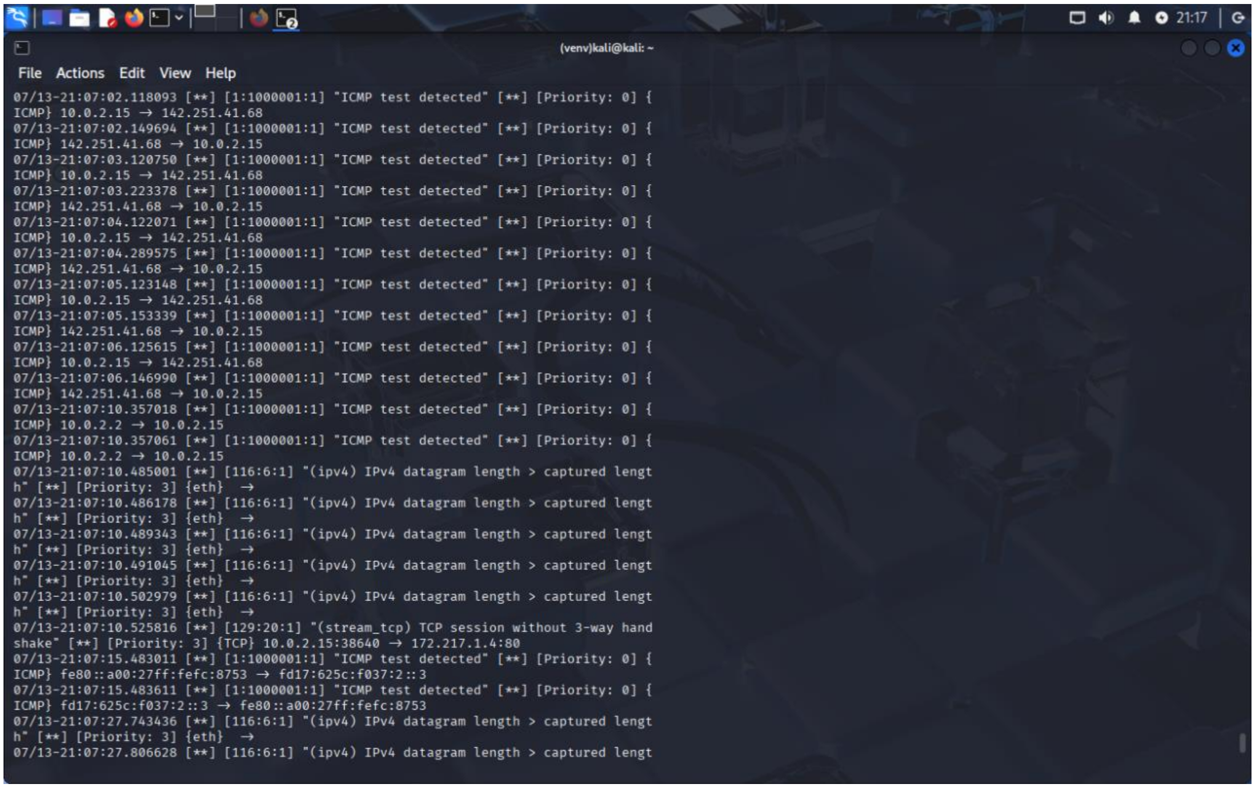
```
gth" [**] [Priority: 3] {eth} →
07/13-20:36:13.239777 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth} →
07/13-20:36:13.240427 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth} →
07/13-20:36:13.256401 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth} →
07/13-20:36:13.259572 [**] [116:275:1] "(ipv6) IPv6 datagram length > captured len
gth" [**] [Priority: 3] {eth} →
07/13-20:36:18.223766 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP} →
07/13-20:36:53.526302 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP} →
07/13-20:41:21.306103 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:43:23.664419 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP} →
07/13-20:50:46.892999 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:50:56.022681 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:50:56.622340 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-20:50:56.770227 [**] [129:20:1] "(stream_tcp) TCP session without 3-way hand
shake" [**] [Priority: 3] {TCP} 10.0.2.15:37284 → 142.251.32.74:443
07/13-20:51:01.673229 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-20:51:01.674478 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
07/13-20:51:01.675132 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP} →
07/13-20:51:20.146680 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:20.322429 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:20.346500 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:20.529284 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:20.942678 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:20.954071 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
```

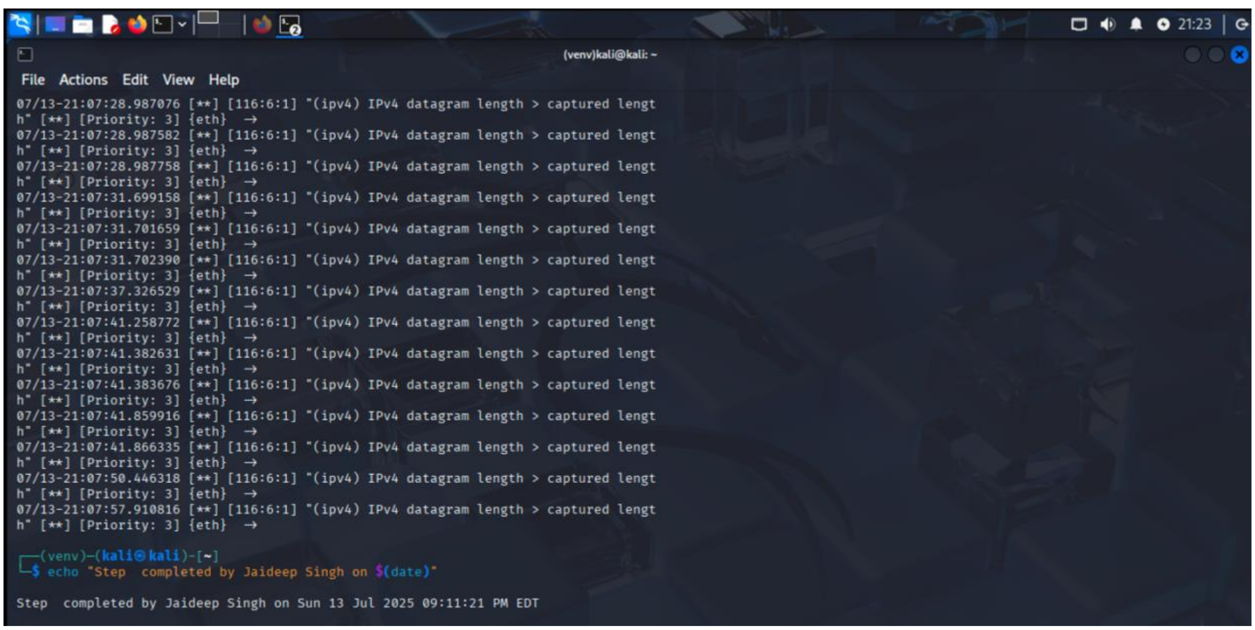

21:17
(venv)kali@kali: ~
File Actions Edit View Help

```
h" [**] [Priority: 3] {eth} →
07/13-20:51:23.702026 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:24.289029 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:24.296997 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:25.981819 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:25.991578 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:27.078144 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:27.078512 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:29.169869 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:29.171284 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:29.174236 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:34.828530 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:34.887945 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:36.502668 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:36.504982 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:36.510594 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:36.513795 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:50.325759 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:54.313313 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-20:51:58.129304 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 8.8.8.8
07/13-20:51:58.152863 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 8.8.8.8 → 10.0.2.15
07/13-20:51:59.131051 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 8.8.8.8
```


21:17
(venv)kali@kali: ~
File Actions Edit View Help
07/13-21:04:27.816941 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 8.8.8.8 → 10.0.2.15
07/13-21:04:31.290726 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:04:37.917035 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-21:04:38.063498 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:04:43.156609 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-21:04:43.157526 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP} →
07/13-21:04:43.157770 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
07/13-21:04:46.837683 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:05.703304 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:05.770324 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:05.870220 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:05.896049 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:05.910645 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:07.087748 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:07.088243 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:11.865669 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:11.866206 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:11.868927 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:17.682199 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:18.285033 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:18.286021 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt
h" [**] [Priority: 3] {eth} →
07/13-21:05:18.291355 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt


21:17
(venv)kali@kali: ~
File Actions Edit View Help
07/13-21:06:43.055050 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-21:06:43.055110 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-21:06:43.080198 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:06:43.099902 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:06:43.101023 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.2 → 10.0.2.15
07/13-21:06:44.081096 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:06:44.123896 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:06:45.082587 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:06:45.105075 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:06:46.085044 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:06:46.111936 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:06:47.088071 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:06:47.111117 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:06:48.080706 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-21:06:48.081400 [**] [112:1:1] "(arp_spoof) unicast ARP request" [**] [Prior
ity: 3] {ARP} →
07/13-21:06:48.081527 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
07/13-21:06:48.088512 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:06:48.110081 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:06:49.090751 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:06:49.114588 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:06:50.092849 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {
ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:06:50.117181 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {


21:17
(venv)kali@kali: ~
File Actions Edit View Help
07/13-21:07:02.118093 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:07:02.149694 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:07:03.120750 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:07:03.223378 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:07:04.122071 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:07:04.289575 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:07:05.123148 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:07:05.153339 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:07:06.125615 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 10.0.2.15 → 142.251.41.68
07/13-21:07:06.146990 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 142.251.41.68 → 10.0.2.15
07/13-21:07:10.357018 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 10.0.2.2 → 10.0.2.15
07/13-21:07:10.357061 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} 10.0.2.2 → 10.0.2.15
07/13-21:07:10.485001 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:10.486178 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:10.489343 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:10.491045 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:10.502979 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:10.525816 [**] [129:20:1] "(stream_tcp) TCP session without 3-way handshake" [**] [Priority: 3] {TCP} 10.0.2.15:38640 → 172.217.1.4:80
07/13-21:07:15.483011 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} fe80::a00:27ff:fefc:8753 → fd17:625c:f037:2::3
07/13-21:07:15.483611 [**] [1:1000001:1] "ICMP test detected" [**] [Priority: 0] {ICMP} fd17:625c:f037:2::3 → fe80::a00:27ff:fefc:8753
07/13-21:07:27.743436 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:27.806628 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured lengt


21:23
(venv)kali@kali: ~
File Actions Edit View Help
07/13-21:07:28.987076 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:28.987582 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:28.987758 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:31.699158 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:31.701659 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:31.702390 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:37.326529 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:41.258772 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:41.382631 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:41.383676 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:41.859916 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:41.866335 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:50.446318 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
07/13-21:07:57.910816 [**] [116:6:1] "(ipv4) IPv4 datagram length > captured length" [**] [Priority: 3] {eth} →
┌──(venv)-(kali㉿kali)-[~]
└─$ echo "Step completed by Jaideep Singh on $(date)"
Step completed by Jaideep Singh on Sun 13 Jul 2025 09:11:21 PM EDT